

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

- 1 presented to TSG for information;
- 2 presented to TSG for approval;
- 3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document establishes the minimum RF requirements for NR User Equipment (UE) operating on frequency Range 1.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

References are either specific (identified by date of publication, edition number, version number, etc.) or nonspecific.

For a specific reference, subsequent revisions do not apply.

For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone".
- [3] 3GPP TS 38.101-3: "NR; User Equipment (UE) radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios".
- [4] 3GPP TS 38.521-1: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Range 1 Standalone".
- [5] Recommendation ITU-R M.1545: "Measurement uncertainty as it applies to test limits for the terrestrial component of International Mobile Telecommunications-2000".
- [6] 3GPP TS 38.211: "NR; Physical channels and modulation".
- [7] 3GPP TS 38.331: "Radio Resource Control (RRC) protocol specification".
- [8] 3GPP TS 38.213: "NR; Physical layer procedures for control".
- [9] ITU-R Recommendation SM.329-10, "Unwanted emissions in the spurious domain".
- [10] 3GPP TS 38.214: "NR; Physical layer procedures for data".
- [11] 3GPP TS 36.101: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception;

- [12] ETSI TS 102 792: "Intelligent Transport Systems (ITS); Mitigation techniques to avoid interference between European CEN Dedicated Short Range Communication (CEN DSRC) equipment and Intelligent Transport Systems (ITS) operating in the 5 GHz frequency range".
- [13] 3GPP TS 38.133: "NR; Requirements for support of radio resource management".
- [14] 3GPP TS 37.213: "Physical layer procedures for shared spectrum channel access".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Aggregated Allocation Bandwidth: Total bandwidth of all allocated RBs in a transmission occasion. Can be calculated for two aggregated CCs as $L_{\text{CRB},1} * 12 * \text{SCS}_1 + L_{\text{CRB},2} * 12 * \text{SCS}_2$.

Aggregated Channel Bandwidth: The RF bandwidth in which a UE transmits and receives multiple contiguously aggregated carriers.

Carrier aggregation: Aggregation of two or more component carriers in order to support wider transmission bandwidths.

Carrier aggregation band: A set of one or more operating bands across which multiple carriers are aggregated with a specific set of technical requirements.

Carrier aggregation bandwidth class: A class defined by the aggregated transmission bandwidth configuration and maximum number of component carriers supported by a UE.

Carrier aggregation configuration: A combination of CA operating band(s) and CA bandwidth class(es) supported by a UE.

Concurrent operation: The simultaneous transmission and reception of sidelink and Uu interfaces while operation is agnostic of the service used on each interface.

Contiguous carriers: A set of two or more carriers configured in a spectrum block where there are no RF requirements based on co-existence for un-coordinated operation within the spectrum block.

Contiguous resource allocation: A resource allocation of consecutive resource blocks within one carrier or across contiguously aggregated carriers. The gap between contiguously aggregated carriers due to the nominal channel spacing is allowed.

Contiguous spectrum: Spectrum consisting of a contiguous block of spectrum with no sub-block gaps.

Inter-band carrier aggregation: Carrier aggregation of component carriers in different operating bands.

NOTE: Carriers aggregated in each band can be contiguous or non-contiguous.

Intra-band contiguous carrier aggregation: Contiguous carriers aggregated in the same operating band.

Intra-band non-contiguous carrier aggregation: Non-contiguous carriers aggregated in the same operating band.

Sub-band: For a UE that supports shared spectrum channel access in wideband operation, a sub-band is the set of RBs within an approximately 20 MHz segment of the channel where the wideband channel is uniformly divided into an integer number of 20 MHz sub-bands. Sub-bands may be separately allocated in uplink and downlink.

Sub-block: This is one contiguous allocated block of spectrum for transmission and reception by the same UE. There may be multiple instances of sub-blocks within an RF bandwidth.

Sub-block bandwidth: The bandwidth of one sub-block.

Sub-block gap: A frequency gap between two consecutive sub-blocks within an RF bandwidth, where the RF requirements in the gap are based on co-existence for un-coordinated operation.

UE transmission bandwidth configuration: Set of resource blocks located within the UE channel bandwidth which may be used for transmitting or receiving by the UE.

Vehicular UE: A UE embedded in a vehicle, permanently connected to an embedded antenna system that radiates externally for NR operating bands.

NOTE: Vehicular UE does not refer to other UE form factors placed inside the vehicle.

Wideband operation: For a UE that supports shared spectrum channel access, wideband operation refers to operation within a channel larger than 20 MHz in which intra-cell guard bands may be configured to distinguish individual RB-sets

3.2 Symbols

For the purposes of the present document, the following symbols apply:

ΔF_{Global}	Granularity of the global frequency raster
ΔF_{Raster}	Band dependent channel raster granularity
Δf_{OOB}	Δ Frequency of Out Of Band emission
$\Delta F_{\text{TX-RX}}$	Δ Frequency of default TX-RX separation of the FDD <i>operating band</i>
ΔMPR_c	Allowed Maximum Power Reduction relaxation for serving cell <i>c</i>
$\Delta P_{\text{PowerClass}}$	Adjustment to maximum output power for a given power class
$F_{\text{offset}}^{\text{RB}}$	The starting frequency offset between the allocated RB and the measured non-allocated RB
$\Delta R_{\text{IB},c}$	Allowed reference sensitivity relaxation due to support for inter-band CA operation, for serving cell <i>c</i>
ΔR_{IBC}	Allowed reference sensitivity relaxation due to support for intra-band contiguous CA operation
ΔR_{IBNC}	Allowed reference sensitivity relaxation due to support for intra-band non-contiguous CA operation
$\Delta R_{\text{IB},4R}$	Reference sensitivity adjustment due to support for 4 antenna ports
Δ_{shift}	Channel raster offset
$F_{\text{offset}}^{\text{C}}$	Allowed operating band edge transmission power relaxation

$\overline{P_{\text{e}}}_{\text{C},c}$	Allowed operating band edge transmission power relaxation for serving cell c
$\Delta T_{\text{IB},c}$	Allowed maximum configured output power relaxation due to support for inter-band CA operation, inter-band NR-DC operation and due to support for SUL operations, for serving cell c
$\text{BW}_{\text{Channel}}$	Channel bandwidth
$\text{BW}_{\text{Channel},\text{block}}$	Sub-block bandwidth, expressed in MHz. $\text{BW}_{\text{Channel},\text{block}} = F_{\text{edge},\text{block},\text{high}} - F_{\text{edge},\text{block},\text{low}}$
$\text{BW}_{\text{Channel_CA}}$	Aggregated channel bandwidth, expressed in MHz
$\text{BW}_{\text{Channel},\text{max}}$	Maximum channel bandwidth supported among all bands in a release
BW_{GB}	$\max(\text{BW}_{\text{GB},\text{Channel}(k)})$
$\text{BW}_{\text{GB},\text{Channel}(k)}$	Minimum guard band defined in clause 5.3A.3 of carrier k
BW_{DL}	Channel bandwidth for DL
BW_{UL}	Channel bandwidth for UL
$\text{BW}_{\text{interferer}}$	Bandwidth of the interferer
$\text{Ceil}(x)$	Rounding upwards; $\text{ceil}(x)$ is the smallest integer such that $\text{ceil}(x) \geq x$
$\text{Floor}(x)$	Rounding downwards; $\text{floor}(x)$ is the greatest integer such that $\text{floor}(x) \leq x$
F_{C}	Center frequency of a carrier for a numerology defined by the <i>RF reference frequency</i> on the channel raster mapped to the carrier according to sub-clause 5.4.2.2
$F_{\text{C},\text{block},\text{high}}$	F_{C} of the highest transmitted/received carrier in a <i>sub-block</i>
$F_{\text{C},\text{block},\text{low}}$	F_{C} of the lowest transmitted/received carrier in a <i>sub-block</i>
$F_{\text{C},\text{low}}$	The F_{C} of the lowest carrier, expressed in MHz
$F_{\text{C},\text{high}}$	The F_{C} of the highest carrier, expressed in MHz
$F_{\text{DL_low}}$	The lowest frequency of the downlink <i>operating band</i>
$F_{\text{DL_high}}$	The highest frequency of the downlink <i>operating band</i>
$F_{\text{UL_low}}$	The lowest frequency of the uplink <i>operating band</i>
$F_{\text{UL_high}}$	The highest frequency of the uplink <i>operating band</i>
$F_{\text{edge},\text{block},\text{low}}$	The lower <i>sub-block</i> edge, where $F_{\text{edge},\text{block},\text{low}} = F_{\text{C},\text{block},\text{low}} - F_{\text{offset},\text{low}}$.
$F_{\text{edge},\text{block},\text{high}}$	The upper <i>sub-block</i> edge, where $F_{\text{edge},\text{block},\text{high}} = F_{\text{C},\text{block},\text{high}} + F_{\text{offset},\text{high}}$.
$F_{\text{edge},\text{low}}$	The <i>lower edge of aggregated channel bandwidth</i> , expressed in MHz. $F_{\text{edge},\text{low}} = F_{\text{C},\text{low}} - F_{\text{offset},\text{low}}$.
$F_{\text{edge},\text{high}}$	The <i>higher edge of aggregated channel bandwidth</i> , expressed in MHz. $F_{\text{edge},\text{high}} = F_{\text{C},\text{high}} + F_{\text{offset},\text{high}}$.
$F_{\text{Interferer}}(\text{offset})$	Frequency offset of the interferer (between the center frequency of the interferer and the carrier frequency of the carrier measured). For intra-band contiguous CA, the $F_{\text{Interferer}}(\text{offset})$ is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer
$F_{\text{Interferer}}$	Frequency of the interferer

F_{Ioffset}	Frequency offset of the interferer (between the center frequency of the interferer and the closest edge of the carrier measured)
F_{offset}	Frequency offset from $F_{\text{C,high}}$ to the <i>higher edge</i> or $F_{\text{C,low}}$ to the <i>lower edge</i> .
$F_{\text{offset,high}}$	Frequency offset from $F_{\text{C,high}}$ to the upper <i>UE RF Bandwidth edge</i> , or from $F_{\text{C,block, high}}$ to the upper sub-block edge
$F_{\text{offset,low}}$	Frequency offset from $F_{\text{C,low}}$ to the lower <i>UE RF Bandwidth edge</i> , or from $F_{\text{C,block, low}}$ to the lower sub-block edge
F_{OOB}	The boundary between the NR out of band emission and spurious emission domains
F_{REF}	RF reference frequency
$F_{\text{REF-Offs}}$	Offset used for calculating F_{REF}
$F_{\text{REF, shift}}$	RF reference frequency for Supplementary Uplink (SUL) bands, the uplink of all FDD bands, and TDD bands
$F_{\text{uw}} (\text{offset})$	The frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer
$\text{GB}_{\text{Channel}}$	Minimum guard band defined in clause 5.3.3, expressed in kHz
L_{CRB}	Transmission bandwidth which represents the length of a contiguous resource block allocation expressed in units of resources blocks
Max()	The largest of given numbers
Min()	The smallest of given numbers
	Physical resource block number
NR_{ACLR}	NR ACLR
NR_{B}	Transmission bandwidth configuration, expressed in units of resource blocks
$\text{NR}_{\text{B_agg}}$	The number of the aggregated RBs within the fully allocated aggregated channel bandwidth for carrier 1 to j, where μ is defined in TS 38.211 [6]
$\text{NR}_{\text{B,c}}$	The transmission bandwidth configuration of component carrier c, expressed in units of resource blocks for carrier j, where μ is defined in TS 38.211 [6]
$\text{NR}_{\text{B,largest BW}}$	The largest transmission bandwidth configuration of the component carriers in the bandwidth combination, expressed in units of resource blocks
$\text{NR}_{\text{B,low}}$	The transmission bandwidth configurations according to Table 5.3.2-1 for the lowest assigned component carrier in clause 5.3A.3
$\text{NR}_{\text{B,high}}$	The transmission bandwidth configurations according to Table 5.3.2-1 for the highest assigned component carrier in clause 5.3A.3
N_{REF}	NR Absolute Radio Frequency Channel Number (NR-ARFCN)
$\text{N}_{\text{REF-Offs}}$	Offset used for calculating N_{REF}
P_{CMAX}	The configured maximum UE output power
$P_{\text{CMAX}, c}$	The configured maximum UE output power for serving cell c
$P_{\text{CMAX}, f, c}$	The configured maximum UE output power for carrier f of serving cell c in each slot

P_{EMAX}	Maximum allowed UE output power signalled by higher layers
$P_{\text{EMAX}, c}$	Maximum allowed UE output power signalled by higher layers for serving cell c
$P_{\text{Interferer}}$	Modulated mean power of the interferer
$P_{\text{largest BW}}$	Power of the largest transmission bandwidth configuration of the component carriers in the bandwidth combination
$P_{\text{PowerClass}}$	The nominal UE power (i.e., no tolerance)
$P\text{-MPR}_c$	Power Management Maximum Power Reduction for serving cell c
P_{RB}	The transmitted power per allocated RB, measured in dBm
P_{UMAX}	The measured configured maximum UE output power
P_{uw}	Power of an unwanted DL signal
P_{w}	Power of a wanted DL signal
RB_{start}	The lowest RB index of transmitted resource blocks
$\text{RB}_{\text{start_CA}}$	The lowest RB index of transmitted resource blocks for intra-band contiguous CA
SCS_c	SCS for the component carrier c , expressed in kHz
$\text{SCS}_{\text{largest BW}}$	SCS for the largest transmission bandwidth configuration of the component carriers in the bandwidth combination, expressed in kHz
SCS_{low}	SCS for the lowest assigned component carrier in clause 5.3A.3, expressed in kHz
SCS_{high}	SCS for the highest assigned component carrier in clause 5.3A.3, expressed in kHz
tp	Transient Period value signalled by the UE
tp_{start}	Start position of transient period relative to the symbol boundary
$T(P_{\text{CMAX}, f, c})$	Tolerance for applicable values of $P_{\text{CMAX}, f, c}$ for configured maximum UE output power for carrier f of serving cell c
$T_{\text{L}, c}$	Absolute value of the lower tolerance for the applicable <i>operating band</i> as specified in clause 6.2.1
SS_{REF}	SS block reference frequency position
$\text{UTRA}_{\text{ACLR}}$	UTRA ACLR

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
A-MPR	Additional Maximum Power Reduction
BS	Base Station

BW	Bandwidth
BWP	Bandwidth Part
CA	Carrier Aggregation
CA_nX-nY	Inter-band CA of component carrier(s) in one sub-block within Band nX and component carrier(s) in one sub-block within Band nY where nX and nY are the applicable NR <i>operating bands</i>
CC	Component Carriers
CG	Carrier Group
CP-OFDM	Cyclic Prefix-OFDM
CW	Continuous Wave
DC	Dual Connectivity
DFT-s-OFDM	Discrete Fourier Transform-spread-OFDM
DM-RS	Demodulation Reference Signal
DTX	Discontinuous Transmission
E-UTRA	Evolved UTRA
EIRP	Equivalent Isotropically Radiated Power
EVM	Error Vector Magnitude
FR	Frequency Range
FRC	Fixed Reference Channel
FWA	Fixed Wireless Access
GSCN	Global Synchronization Channel Number
IBB	In-band Blocking
IDFT	Inverse Discrete Fourier Transformation
ITS	Intelligent Transportation System
ITUR	Radiocommunication Sector of the International Telecommunication Union
MBW	Measurement bandwidth
MCG	Master Cell Group
MOP	Maximum Output Power
MPR	Allowed maximum power reduction
MSD	Maximum Sensitivity Degradation
NR	New Radio
NR-ARFCN	NR Absolute Radio Frequency Channel Number
NS	Network Signalling
OCNG	OFDMA Channel Noise Generator
OOB	Out-of-band

P-MPR	Power Management Maximum Power Reduction
PRB	Physical Resource Block
PSCCH	Physical Sidelink Control CHannel
PSSCH	Physical Sidelink Shared CHannel
QAM	Quadrature Amplitude Modulation
RE	Resource Element
REFSENS	Reference Sensitivity
RF	Radio Frequency
RMS	Root Mean Square (value)
RSRP	Reference Signal Receiving Power
Rx	Receiver
SC	Single Carrier
SCG	Secondary Cell Group
SCS	Subcarrier spacing
SDL	Supplementary Downlink
SEM	Spectrum Emission Mask
SL	Sidelink
SL-MIMO	Sidelink-Multiple Antenna transmission
SNR	Signal-to-Noise Ratio
SRS	Sounding Reference Symbol
SS	Synchronization Symbol
SUL	Supplementary uplink
TAE	Time Alignment Error
TAG	Timing Advance Group
Tx	Transmitter
UL MIMO	Uplink Multiple Antenna transmission
ULFPTx	Uplink Full Power Transmission
V2X	Vehicle to Everything

4 General

4.1 Relationship between minimum requirements and test requirements

The present document is a Single-RAT specification for NR UE, covering RF characteristics and minimum performance requirements. Conformance to the present specification is demonstrated by fulfilling the test requirements specified in the conformance specification 3GPP TS 38.521-1 [4].

The Minimum Requirements given in this specification make no allowance for measurement uncertainty. The test specification TS 38.521-1 [4] defines test tolerances. These test tolerances are individually calculated for each test. The test tolerances are used to relax the minimum requirements in this specification to create test requirements. For some requirements, including regulatory requirements, the test tolerance is set to zero.

The measurement results returned by the test system are compared - without any modification - against the test requirements as defined by the shared risk principle.

The shared risk principle is defined in Recommendation ITUR M.1545 [5].

4.2 Applicability of minimum requirements

- a) In this specification the Minimum Requirements are specified as general requirements and additional requirements. Where the Requirement is specified as a general requirement, the requirement is mandated to be met in all scenarios
- b) For specific scenarios for which an additional requirement is specified, in addition to meeting the general requirement, the UE is mandated to meet the additional requirements.
- c) The spurious emissions power requirements are for the long-term average of the power. For the purpose of reducing measurement uncertainty it is acceptable to average the measured power over a period of time sufficient to reduce the uncertainty due to the statistical nature of the signal
- d) All the requirements for intra-band contiguous and non-contiguous CA apply under the assumption of the same slot format indicated by *TDD-UL-DL-ConfigurationCommon* and *TDD-UL-DL-ConfigurationDedicated* in the PCell and SCells for NR SA.

4.3 Specification suffix information

Unless stated otherwise the following suffixes are used for indicating at 2nd level clause, shown in Table 4.3-1.

Table 4.3-1: Definition of suffixes

Clause suffix	Variant
None	Single Carrier
A	Carrier Aggregation (CA)
B	Dual-Connectivity (DC)
C	Supplementary Uplink (SUL)

D	UL MIMO
E	V2X
F	Shared spectrum channel access

A terminal which supports the above features needs to meet both the general requirements and the additional requirement applicable to the additional clause (suffixes A to F) in clauses 5, 6 and 7. Where there is a difference in requirement between the general requirements and the additional clause requirements (suffixes A to F) in clauses 5, 6 and 7, the tighter requirements are applicable unless stated otherwise in the additional clause.

A terminal which supports more than one feature in clauses 5, 6 and 7 shall meet all of the separate corresponding requirements.

For a terminal that supports SUL for the band combination specified in Table 5.2C-1, the current version of the specification assumes the terminal is configured with active transmission either on UL carrier or SUL carrier at any time in one serving cell and the UE requirements for single carrier shall apply for the active UL or SUL carrier accordingly.

For a terminal that supports SUL band combinations specified in Table 5.2C-2, the current version of the specification assumes the terminal is configured with active transmission either on UL carrier or SUL carrier at any time, and the UE requirements for the active CA configuration or SUL carrier shall apply accordingly.

For a terminal that supports SUL, the current version of the specification assumes the terminal is not configured with UL MIMO on SUL carrier.

For a terminal that supports operation in shared spectrum, the current version of this specification assumes in the uplink sub-bands within a wideband channel shall be contiguously allocated to the UE. The uplink requirements for one or more non-transmitted sub-bands between two transmitted sub-bands does not form a part of the current version of this specification.

5 Operating bands and channel arrangement

5.1 General

The channel arrangements presented in this clause are based on the operating bands and channel bandwidths defined in the present release of specifications.

NOTE: Other operating bands and channel bandwidths may be considered in future releases.

Requirements throughout the RF specifications are in many cases defined separately for different frequency ranges (FR). The frequency ranges in which NR can operate according to this version of the specification are identified as described in Table 5.1-1.

Table 5.1-1: Definition of frequency ranges

Frequency range designation	Corresponding frequency range
FR1	410 MHz – 7125 MHz

FR2	24250 MHz – 52600 MHz
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The present specification covers FR1 operating bands.

5.2 Operating bands

NR is designed to operate in the FR1 operating bands defined in Table 5.2-1.

Table 5.2-1: NR operating bands in FR1

NR operating band	Uplink (UL) <i>operating band</i> BS receive / UE transmit	Downlink (DL) <i>operating band</i> BS transmit / UE receive	Duplex Mode
	F_{UL_low} – F_{UL_high}	F_{DL_low} – F_{DL_high}	
n1	1920 MHz – 1980 MHz	2110 MHz – 2170 MHz	FDD
n2	1850 MHz – 1910 MHz	1930 MHz – 1990 MHz	FDD
n3	1710 MHz – 1785 MHz	1805 MHz – 1880 MHz	FDD
n5	824 MHz – 849 MHz	869 MHz – 894 MHz	FDD
n7	2500 MHz – 2570 MHz	2620 MHz – 2690 MHz	FDD
n8	880 MHz – 915 MHz	925 MHz – 960 MHz	FDD
n12	699 MHz – 716 MHz	729 MHz – 746 MHz	FDD
n14	788 MHz – 798 MHz	758 MHz – 768 MHz	FDD
n18	815 MHz – 830 MHz	860 MHz – 875 MHz	FDD
n20	832 MHz – 862 MHz	791 MHz – 821 MHz	FDD
n25	1850 MHz – 1915 MHz	1930 MHz – 1995 MHz	FDD
n26	814 MHz – 849 MHz	859 MHz – 894 MHz	FDD
n28	703 MHz – 748 MHz	758 MHz – 803 MHz	FDD
n29 ¹⁵	N/A	717 MHz – 728 MHz	SDL
n30 ³	2305 MHz – 2315 MHz	2350 MHz – 2360 MHz	FDD
n34	2010 MHz – 2025 MHz	2010 MHz – 2025 MHz	TDD
n38 ¹⁰	2570 MHz – 2620 MHz	2570 MHz – 2620 MHz	TDD
n39	1880 MHz – 1920 MHz	1880 MHz – 1920 MHz	TDD
n40	2300 MHz – 2400 MHz	2300 MHz – 2400 MHz	TDD
n41	2496 MHz – 2690 MHz	2496 MHz – 2690 MHz	TDD
n46 ^{13,14}	5150 MHz – 5925 MHz	5150 MHz – 5925 MHz	TDD
n47 ¹¹	5855 MHz – 5925 MHz	5855 MHz – 5925 MHz	TDD
n48	3550 MHz – 3700 MHz	3550 MHz – 3700 MHz	TDD
n50 ¹	1432 MHz – 1517 MHz	1432 MHz – 1517 MHz	TDD
n51	1427 MHz – 1432 MHz	1427 MHz – 1432 MHz	TDD

n53	2483.5 MHz – 2495 MHz	2483.5 MHz – 2495 MHz	TDD
n65 ⁴	1920 MHz – 2010 MHz	2110 MHz – 2200 MHz	FDD
n66	1710 MHz – 1780 MHz	2110 MHz – 2200 MHz	FDD ^{6,7}
n70	1695 MHz – 1710 MHz	1995 MHz – 2020 MHz	FDD
n71	663 MHz – 698 MHz	617 MHz – 652 MHz	FDD
n74	1427 MHz – 1470 MHz	1475 MHz – 1518 MHz	FDD
n75 ¹⁵	N/A	1432 MHz – 1517 MHz	SDL ²
n76 ¹⁵	N/A	1427 MHz – 1432 MHz	SDL
n77 ¹²	3300 MHz – 4200 MHz	3300 MHz – 4200 MHz	TDD
n78	3300 MHz – 3800 MHz	3300 MHz – 3800 MHz	TDD
n79	4400 MHz – 5000 MHz	4400 MHz – 5000 MHz	TDD
n80	1710 MHz – 1785 MHz	N/A	SUL
n81	880 MHz – 915 MHz	N/A	SUL
n82	832 MHz – 862 MHz	N/A	SUL
n83	703 MHz – 748 MHz	N/A	SUL
n84	1920 MHz – 1980 MHz	N/A	SUL
n86	1710 MHz – 1780 MHz	N/A	SUL
n89	824 MHz – 849 MHz	N/A	SUL
n90 ⁵	2496 MHz – 2690 MHz	2496 MHz – 2690 MHz	TDD
n91 ⁹	832 MHz – 862 MHz	1427 MHz – 1432 MHz	FDD
n92 ⁹	832 MHz – 862 MHz	1432 MHz – 1517 MHz	FDD
n93 ⁹	880 MHz – 915 MHz	1427 MHz – 1432 MHz	FDD
n94 ⁹	880 MHz – 915 MHz	1432 MHz – 1517 MHz	FDD
n95 ⁸	2010 MHz – 2025 MHz	N/A	SUL
n96 ^{13,14}	5925 MHz – 7125 MHz	5925 MHz – 7125 MHz	TDD

NOTE 1: UE that complies with the NR Band n50 minimum requirements in this specification shall also comply with the NR Band n51 minimum requirements.

NOTE 2: UE that complies with the NR Band n75 minimum requirements in this specification shall also comply with the NR Band n76 minimum requirements.

NOTE 3: Uplink transmission is not allowed at this band for UE with external vehicle-mounted antennas.

NOTE 4: A UE that complies with the NR Band n65 minimum requirements in this specification shall also comply with the NR Band n1 minimum requirements.

NOTE 5: Unless otherwise stated, the applicability of requirements for Band n90 is in accordance with that for Band n41; a UE supporting Band n90 shall meet the requirements for Band n41. A UE supporting Band n90 shall also support band n41.

NOTE 6: A UE that supports NR Band n66 shall receive in the entire DL operating band.

NOTE 7: A UE that supports NR Band n66 and CA operation in any CA band shall also comply with the minimum requirements specified for the DL CA configurations CA_n66B and CA_n66(2A) in the current version of the specification.

NOTE 8: This band is applicable in China only.
NOTE 9: Variable duplex operation does not enable dynamic variable duplex configuration by the network, and is used such that DL and UL frequency ranges are supported independently in any valid frequency range for the band.
NOTE 10: When this band is used for V2X SL service, the band is exclusively used for NR V2X in particular regions.
NOTE 11: This band is unlicensed band used for V2X service. There is no expected network deployment in this band.
NOTE 12: In the USA this band is restricted to 3450 – 3550 MHz and 3700 – 3980 MHz.
NOTE 13: This band is restricted to operation with shared spectrum channel access as defined in TS 37.213 [14].
NOTE 14: This band is applicable only in countries/regions designating this band for shared-spectrum access use subject to country-specific conditions.
NOTE 15: For SDL bands, downlink configuration for RRM performance testing is same as FDD.

5.2A Operating bands for CA

5.2A.0 General

CA operating bands including Band n90 are defined by the corresponding CA operating bands including Band n41 with Band n90 replacing Band n41. For brevity the said CA operating bands including Band n90 are not listed in the tables below but are covered by this specification.

5.2A.1 Intra-band CA

NR intra-band carrier aggregation is designed to operate in the operating bands defined in Table 5.2A.1-1 and Table 5.2A.1-2, where all operating bands are within FR1.

Table 5.2A.1-1: Intra-band contiguous CA operating bands in FR1

NR CA Band	NR Band (Table 5.2-1)
CA_n1	n1
CA_n7	n7
CA_n40	n40
CA_n41	n41
CA_n46	n46
CA_n48	n48
CA_n66	n66
CA_n71	n71

CA_n77	n77
CA_n78	n78
CA_n79	n79
NOTE 1: The minimum requirements only apply for non simultaneous Tx/Rx between all carriers for TDD combinations.	

Table 5.2A.1-2: Intra-band non-contiguous CA operating bands in FR1

NR CA Band	NR Band (Table 5.2-1)
CA_n3(*)	n3
CA_n7(*)	n7
CA_n25(*)	n25
CA_n41(*)	n41
CA_n48(*)	n48
CA_n66(*)	n66
CA_n77(*)	n77
CA_n78(*)	n78
NOTE 1: The minimum requirements only apply for non simultaneous Tx/Rx between all carriers for TDD combinations.	
NOTE 2: The notation CA_nX(*) in this table indicates intra-band non-contiguous CA for band nX. The configurations for each band are in 5.5A.2.	

5.2A.2 Inter-band CA

NR inter-band carrier aggregation is designed to operate in the operating bands defined in Table 5.2A.2.1-1, Table 5.2A.2.2-1 and Table 5.2A.2.3-1, where all operating bands are within FR1.

If the mandatory simultaneous Rx/Tx capability applies for a lower order band combination, when the applicable lower order band combination is a band pair in a higher order band combination, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order band combination.

Table 5.2A.2-1: Void

Table 5.2A.2-2: Void

Table 5.2A.2-3: Void

5.2A.2.1 Inter-band CA (two bands)

Table 5.2A.2.1-1: Inter-band CA operating bands involving FR1 (two bands)

NR CA Band	NR Band (Table 5.2-1)	DL interruption allowed (Note 8)
CA_n1-n3	n1, n3	
CA_n1-n7	n1, n7	
CA_n1-n8	n1, n8	
CA_n1-n28	n1, n28	
CA_n1-n40	n1, n40	
CA_n1-n41 ¹	n1, n41	
CA_n1-n77 ¹	n1, n77	No
CA_n1-n78 ¹	n1, n78	No
CA_n1-n79 ¹	n1, n79	No
CA_n2-n5	n2, n5	
CA_n2-n48	n2, n48	
CA_n2-n66	n2, n66	
CA_n2-n77	n2, n77	
CA_n2-n78	n2, n78	
CA_n3-n7	n3, n7	
CA_n3-n8	n3, n8	
CA_n3-n28	n3, n28	
CA_n3-n38	n3, n38	
CA_n3-n40 ¹	n3, n40	
CA_n3-n41 ¹	n3, n41	No
CA_n3-n77 ¹	n3, n77	No
CA_n3-n78 ¹	n3, n78	No
CA_n3-n79 ¹	n3, n79	No
CA_n5-n7	n5, n7	
CA_n5-n66	n5, n66	
CA_n5-n77 ¹	n5, n77	
CA_n5-n78 ¹	n5, n78	No
CA_n5-n79 ¹	n5, n79	No
CA_n7-n25	n7, n25	
CA_n7-n28	n7, n28	

CA_n7-n66	n7, n66	
CA_n7-n78 ¹	n7, n78	
CA_n8-n39 ¹	n8, n39	
CA_n8-n40 ¹	n8, n40	
CA_n8-n41 ¹	n8, n41	No
CA_n8-n75 ¹	n8, n75	
CA_n8-n78 ¹	n8, n78	No
CA_n8-n79 ¹	n8, n79	No
CA_n20-n28 ²	n20, n28	
CA_n20-n75	n20, n75	
CA_n20-n78	n20, n78	
CA_n25-n41	n25, n41	
CA_n25-n46 ⁶	n25, n46	
CA_n25-n66	n25, n66	
CA_n25-n71	n25, n71	
CA_n25-n78	n25, n78	
CA_n28-n40 ¹	n28, n40	
CA_n28-n41 ¹	n28, n41	
CA_n28-n50	n28, n50	
CA_n28-n75 ²	n28, n75	
CA_n28-n77 ¹	n28, n77	No
CA_n28-n78 ¹	n28, n78	No
CA_n29-n66	n29, n66	
CA_n29-n70	n29, n70	
CA_n38-n66	n38, n66	
CA_n38-n78 ¹	n38, n78	
CA_n39-n40	n39, n40	
CA_n39-n41	n39, n41	No
CA_n39-n79 ¹	n39, n79	No
CA_n40-n41	n40, n41	
CA_n40-n78 ¹	n40, n78	
CA_n40-n79 ^{1,4}	n40, n79	No
CA_n41-n50 ¹	n41, n50	
CA_n41-n66	n41, n66	
CA_n41-n71 ¹	n41, n71	
CA_n41-n78 ¹	n41, n78	
CA_n41-n79 ^{1,3}	n41, n79	No

CA_n46-n48 ⁶	n46, n48	
CA_n46-n66 ⁶	n46, n66	
CA_n48-n66	n48, n66	
CA_n50-n78	n50, n78	
CA_n66-n70	n66, n70	
CA_n66-n71	n66, n71	
CA_n66-n77	n66, n77	
CA_n66-n78	n66, n78	
CA_n70-n71	n70, n71	
CA_n75-n78 ¹	n75, n78	
CA_n76-n78 ¹	n76, n78	
CA_n77-n78 ⁷	n77, n78	
CA_n77-n79 ⁷	n77, n79	
CA_n78-n79 ⁵	n78, n79	
CA_n78-n92	n78, n92	
NOTE 1: Applicable for UE supporting inter-band carrier aggregation with mandatory simultaneous Rx/Tx capability. NOTE 2: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL. NOTE 3: The frequency range below 2506 MHz for Band n41 is not used in this combination. NOTE 4: Applicable for frequency range above 4800 MHz for Band n79 in this combination. NOTE 5: For UEs supporting band n77, the minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n78-n79 NR carriers. This restriction applies also for these carriers when applicable NR CA configuration is part of a higher order configuration. NOTE 6: The PCell is allocated in the licensed band in this combination. NOTE 7: The minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n77-n78 or n77-n79 NR carriers. This restriction applies also for these carriers when applicable NR CA configuration is part of a higher order configuration. NOTE 8: Applicable when dynamic switching between two uplink carriers is conducted. The DL interruption requirement is specified in clause 8.2.2.2.10 of 38.133 [13].		

5.2A.2.2 Inter-band CA (three bands)

Table 5.2A.2.2-1: Inter-band CA operating bands involving FR1 (three bands)

NR CA Band	NR Band (Table 5.2-1)
CA_n1-n3-n7	n1, n3, n7

CA_n1-n3-n8	n1, n3, n8
CA_n1-n3-n28	n1, n3, n28
CA_n1-n3-n41 ³	n1, n3, n41
CA_n1-n3-n78 ³	n1, n3, n78
CA_n1-n7-n28	n1, n7, n28
CA_n1-n7-n78 ³	n1, n7, n78
CA_n1-n8-n78 ³	n1, n8, n78
CA_n1-n28-n78 ³	n1, n28, n78
CA_n1-n40-n78 ³	n1, n40, n78
CA_n3-n7-n28	n3, n7, n28
CA_n3-n7-n78 ³	n3, n7, n78
CA_n3-n8-n78 ³	n3, n8, n78
CA_n3-n28-n77 ³	n3, n28, n77
CA_n3-n28-n78 ³	n3, n28, n78
CA_n3-n40-n41	n3, n40, n41
CA_n3-n41-n79 ³	n3, n41, n79
CA_n5-n66-n78	n5, n66, n78
CA_n7-n25-n66	n7, n25, n66
CA_n7-n28-n78	n7, n28, n78
CA_n7-n66-n78	n7, n66, n78
CA_n8-n39-n41	n8, n39, n41
CA_n8-n41-n79 ³	n8, n41, n79
CA_n20-n28-n78	n20, n28, n78
CA_n25-n41-n66	n25, n41, n66
CA_n25-n41-n71	n41, n66, n71
CA_n25-n66-n71	n25, n66, n71
CA_n25-n66-n78	n25, n66, n78
CA_n28-n40-n78	n28, n40, n78
CA_n28-n41-n78 ³	n28, n41, n78
CA_n29-n66-n70	n29, n66, n70
CA_n39-n41-n79	n39, n41, n79
CA_n40-n41-n79 ^{1,2}	n40, n41, n79
CA_n41-n66-n71	n41, n66, n71
CA_n66-n70-n71	n66, n70, n71

NOTE 1:	The frequency range below 2506 MHz for Band n41 is not used in this band combination.
NOTE 2:	Applicable for frequency range above 4800 MHz for Band n79 in this band combination.
NOTE 3:	Applicable for UE supporting inter-band carrier aggregation with mandatory simultaneous Rx/Tx capability.

5.2A.2.3 Inter-band CA (four bands)

Table 5.2A.2.3-1: Inter-band CA operating bands involving FR1 (four bands)

NR CA Band	NR Band (Table 5.2-1)
CA_n1-n3-n7-n28	n1, n3, n7, n28
CA_n1-n3-n7-n78 ¹	n1, n3, n7, n78
CA_n1-n3-n8-n78 ¹	n1, n3, n8, n78
CA_n1-n3-n28-n78 ¹	n1, n3, n28, n78
CA_n3-n7-n28-n78	n3, n7, n28, n78
CA_n7-n25-n66-n78	n7, n25, n66, n78
NOTE 1: Applicable for UE supporting inter-band carrier aggregation with mandatory simultaneous Rx/Tx capability.	

5.2B Operating bands for DC

The operating bands are specified in clause 5.5B for operation with NR dual connectivity configured, where all operating bands are within FR1.

5.2C Operating band combination for SUL

NR operation is designed to operate in the operating band combination defined in Table 5.2C-1 and Table 5.2C-2, where all operating bands are within FR1.

If the mandatory simultaneous Rx/Tx capability applies for a lower order band combination, when the applicable lower order band combination is a band pair in a higher order band combination, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order band combination.

Table 5.2C-1: Operating band combination for SUL in FR1

NR Band combination for SUL	NR Band (Table 5.2-1)
SUL_n41-n80 ²	n41, n80
SUL_n41-n81 ²	n41, n81
SUL_n41-n95 ²	n41, n95
SUL_n77-n80 ²	n77, n80
SUL_n77-n84 ²	n77, n84
SUL_n78-n80 ²	n78, n80
SUL_n78-n81 ²	n78, n81
SUL_n78-n82 ²	n78, n82
SUL_n78-n83 ²	n78, n83
SUL_n78-n84 ²	n78, n84
SUL_n78-n86 ²	n78, n86
SUL_n79-n80 ²	n79, n80
SUL_n79-n81 ²	n79, n81
SUL_n79-n84 ²	n79, n84
SUL_n79-n95 ²	n79, n95
NOTE 1: If a UE is configured with both NR UL and NR SUL carriers in a cell, the switching time between NR UL carrier and NR SUL carrier is 0 us. NOTE 2: For UE supporting SUL band combination simultaneous Rx/Tx capability is mandatory. NOTE 3: For UE supporting SUL band combination, UL MIMO is not configured on SUL carrier	

Table 5.2C-2: Operating SUL band combination with intra-band non-contiguous CA in FR1

NR Band combination for SUL	NR Band (Table 5.2-1)
CA_n78(*)-n86 ²	n78, n86
NOTE 1: If a UE is configured with both NR UL and NR SUL carriers in a cell, the switching time between NR UL carrier and NR SUL carrier is 0 us. NOTE 2: For UE supporting SUL band combination simultaneous Rx/Tx capability is mandatory. NOTE 3: For UE supporting SUL band combination, UL MIMO is not configured on SUL carrier.	

NOTE 4: The notation CA_nX(*) in this table indicates intra-band non-contiguous CA for band nX. The configurations for each band are in table 5.5C-2.

5.2D Operating bands for UL MIMO

NR is designed to support UL MIMO where all of the operating bands are in FR1 defined in Table 5.2D-1.

Table 5.2D-1: NR operating bands for UL MIMO in FR1

NR operating band
n1
n2
n3
n7
n25
n30 ¹
n34
n38
n39
n40
n41
n46
n48
n66
n70
n71 ²
n77
n78
n79
n96
NOTE 1: Uplink transmission is not allowed at this band for UE with external vehicle-mounted antennas.
NOTE 2: UL MIMO is targeted for FWA form factor.

5.2E Operating band for V2X

5.2E.1 V2X operating bands

NR V2X is designed to operate in the operating bands in FR1 defined in Table 5.2E.1-1.

Table 5.2E.1-1 V2X operating bands in FR1

V2X Operating Band	Sidelink (SL) Transmission operating band	Sidelink (SL) Reception operating band	Duplex Mode	Interface
	$F_{UL_low} - F_{UL_high}$	$F_{DL_low} - F_{DL_high}$		
n38 ¹	2570 MHz - 2620 MHz	2570 MHz - 2620 MHz	HD	PC5
n47	5855 MHz - 5925 MHz	5855 MHz - 5925 MHz	HD	PC5
Note 1: When this band is used for V2X SL service, the band is exclusively used for NR V2X in particular regions.				

5.2E.2 V2X operating bands for concurrent operation

NR V2X operation is designed to operate concurrent with NR uplink/downlink on the operating bands combinations listed in Table 5.2E.2-1.

Table 5.2E.2-1 Inter-band concurrent V2X operating bands

V2X concurrent operating Band	NR or V2X Operating Band	Interface
V2X_n71-n47	n71	Uu
	n47	PC5

5.3 UE channel bandwidth

5.3.1 General

The UE channel bandwidth supports a single NR RF carrier in the uplink or downlink at the UE. From a BS perspective, different UE channel bandwidths may be supported within the same spectrum for transmitting to and receiving from UEs connected to the BS. Transmission of multiple carriers to the same UE (CA) or multiple carriers to different UEs within the BS channel bandwidth can be supported.

From a UE perspective, the UE is configured with one or more BWP / carriers, each with its own UE channel bandwidth. The UE does not need to be aware of the BS channel bandwidth or how the BS allocates bandwidth to different UEs.

The placement of the UE channel bandwidth for each UE carrier is flexible but can only be completely within the BS channel bandwidth.

The relationship between the channel bandwidth, the guardband and the maximum transmission bandwidth configuration is shown in Figure 5.3.1-1.

Figure 5.3.1-1: Definition of the channel bandwidth and the maximum transmission bandwidth configuration for one NR channel

5.3.2 Maximum transmission bandwidth configuration

The maximum transmission bandwidth configuration N_{RB} for each UE channel bandwidth and subcarrier spacing is specified in Table 5.3.2-1.

Table 5.3.2-1: Maximum transmission bandwidth configuration N_{RB}

SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz
	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}	N_{RB}
15	25	52	79	106	133	160	216	270	N/A	N/A	N/A	N/A	N/A
30	11	24	38	51	65	78	106	133	162	189	217	245	273
60	N/A	11	18	24	31	38	51	65	79	93	107	121	135

5.3.3 Minimum guardband and transmission bandwidth configuration

The minimum guardband for each UE channel bandwidth and SCS is specified in Table 5.3.3-1,

Table 5.3.3-1: Minimum guardband for each UE channel bandwidth and SCS (kHz)

SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz
15	242.5	312.5	382.5	452.5	522.5	592.5	552.5	692.5	N/A	N/A	N/A	N/A	N/A
30	505	665	645	805	785	945	905	1045	825	965	925	885	845
60	N/A	1010	990	1330	1310	1290	1610	1570	1530	1490	1450	1410	1370

NOTE: The minimum guardbands have been calculated using the following equation: $GB_{channel} = (BW_{Channel} \times 1000 \text{ (kHz)} - N_{RB} \times SCS \times 12) / 2 - SCS/2$, where N_{RB} are from Table 5.3.2-1 and $GB_{channel}$ expressed in kHz.

Figure 5.3.3-1: Void

The number of RBs configured in any channel bandwidth shall ensure that the minimum guardband specified in this clause is met.

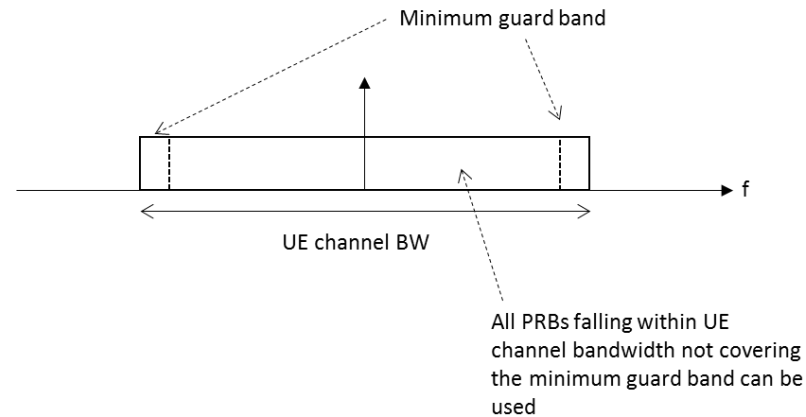


Figure 5.3.3-2: UE PRB utilization

In the case that multiple numerologies are multiplexed in the same symbol due to BS transmission of SSB, the minimum guardband on each side of the carrier is the guardband applied at the configured channel bandwidth for the numerology that is received immediately adjacent to the guard.

If multiple numerologies are multiplexed in the same symbol and the UE channel bandwidth is >50 MHz, the minimum guardband applied adjacent to 15 kHz SCS shall be the same as the minimum guardband defined for 30 kHz SCS for the same UE channel bandwidth.

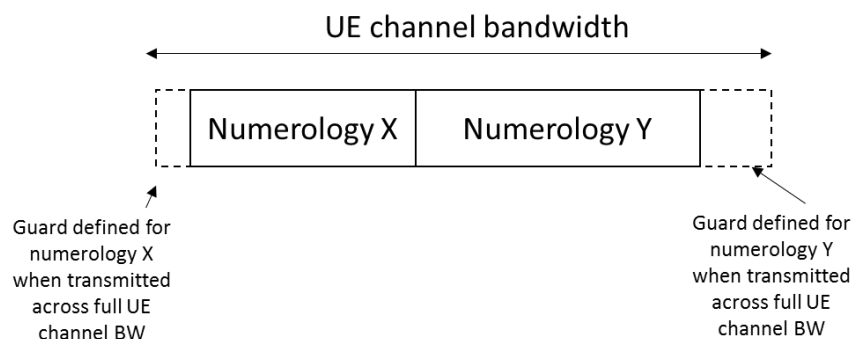


Figure 5.3.3-3 Guard band definition when transmitting multiple numerologies

NOTE: Figure 5.3.3-3 is not intended to imply the size of any guard between the two numerologies. Inter-numerology guard band within the carrier is implementation dependent.

For a UE supporting wideband operation, the nominal intra-cell guard bands and the corresponding sizes of the RB sets separated by the said guard bands are as specified in Table 5.3.3-2 for each UE channel bandwidth and sub-carrier spacing for the downlink and uplink. The nominal intra-cell guard bands in Table 5.3.3-2 are applicable when the respective IE *intraCellGuardBandsUL-List* and *intraCellGuardBandsDL-List* [7] for the uplink and downlink are not provided, as specified in [10] clause 7.

Table 5.3.3-2: Nominal intra-cell guard bands for wideband operation

SCS (kHz)	40 MHz	60 MHz	80 MHz
15	105-6-105 (216)	N/A	N/A
30	50-6-50 (106)	50-6-50-6-50 (162)	50-6-50-5-50-6-50 (217)
60	23-5-23 (51)	23-5-23-5-23 (79)	23-5-23-5-23-5-23 (107)
NOTE 1: The intra-cell guard band is denoted $TBW_0-GB_0-\dots-GB_{N_RBset-2}-TBW_{N_RBset-1}$ for $N_RBset > 1$ number of RB-sets with TBW_i the maximum transmission			

bandwidth (PRB) of RB-set r and GB_r the guard band (PRB) above the upper edge of RB-set r . The RB-set 0 is starting at the first common resource block (CRB) of the carrier as indicated by *offsetToCarrier*. The total transmission bandwidth configuration (size of resource grid) including guard bands is given in between parentheses.

For a UE that supports shared spectrum channel access, there are no uplink or downlink intra-cell guard bands for operation with 10 MHz and 20 MHz channel bandwidths; the maximum transmission bandwidth configurations for these channel bandwidths are in accordance with clause 5.3.2. For each UE channel bandwidth and sub-carrier spacing given by Table 5.3.3-2, the maximum transmission bandwidth configuration of the carrier including intra-cell guard bands, if configured for the uplink and downlink by the respective IE *intraCellGuardBandsUL-List* and *intraCellGuardBandsDL-List* [7], and corresponding RB-set(s) shall be in accordance with clause 5.3.2 with a minimum inter-cell guard band of the UE channel bandwidth as specified in Table 5.3.3-1 for the uplink and downlink. Minimum requirements specified for wideband operation in Clause 6 and Clause 7 also apply for intra-cell guard bands larger than the nominal sizes in Table 5.3.3-2 as listed in Table 5.3.3-3 for each sub-carrier spacing; each guard band in order of CRB index must be larger than or equal to the corresponding nominal guard band specified in Table 5.3.3-2 for each channel bandwidth.

Table 5.3.3-3: Applicable intra-cell guard bands for wideband operation

Parameter	Unit	SCS	
		15 kHz	30 kHz
Intra-cell guard band (size)	PRB	6,7	5,6,7
Transmission bandwidth (size) of RB-set	PRB	104,105	49,50,51

If the UE is configured with zero width intra-cell guard bands for the uplink and downlink by the IE *intraCellGuardBandsUL-List* and *intraCellGuardBandsDL-List* [7] on a carrier greater than 20 MHz, the maximum transmission bandwidth configuration for the uplink and downlink shall be in accordance with clause 5.3.2 with a minimum inter-cell guard band of the UE channel bandwidth as specified in Table 5.3.3-1.

5.3.4 RB alignment

For each numerology, its common resource blocks are specified in Clause 4.4.4.3 in TS 38.211 [6], and the starting point of its transmission bandwidth configuration on the common resource block grid for a given channel bandwidth is indicated by an offset to "Reference point A" in the unit of the numerology. The *UE transmission bandwidth configuration* is indicated by the higher layer parameter *carrierBandwidth* [7] and will fulfil the minimum UE guardband requirement specified in Clause 5.3.3.

5.3.5 UE channel bandwidth per operating band

The requirements in this specification apply to the combination of channel bandwidths, SCS and operating bands shown in Table 5.3.5-1. The transmission bandwidth configuration in Table 5.3.2-1 shall be supported for each of the specified channel bandwidths. The channel bandwidths are specified for both the TX and RX path.

Table 5.3.5-1 Channel bandwidths for each NR band

NR band / SCS / UE Channel bandwidth														
NR Band	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz
n1	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
n2	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60		Yes	Yes	Yes									
n3	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes	Yes						
n5	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n7	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
n8	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n12	15	Yes	Yes	Yes										
	30		Yes	Yes										
	60													
n14	15	Yes	Yes											
	30		Yes											

	60													
n18	15	Yes	Yes	Yes										
	30		Yes	Yes										
	60													
n20	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n25	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes	Yes						
n26	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
n28	15	Yes	Yes	Yes	Yes ⁷		Yes ⁷							
	30		Yes	Yes	Yes ⁷		Yes ⁷							
	60													
n29	15	Yes	Yes											
	30		Yes											
	60													
n30	15	Yes	Yes											
	30		Yes											
	60													
n34	15	Yes	Yes	Yes										
	30		Yes	Yes										
	60		Yes	Yes										
n38 ¹⁰	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes	Yes						
n39	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes	Yes						
n40	15	Yes ⁵	Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		Yes		
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		Yes		
n41	15		Yes	Yes	Yes		Yes	Yes	Yes					

	30		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes	Yes	Yes
	60		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes	Yes	Yes
n46	15		Yes ⁵		Yes			Yes						
	30		Yes ⁵		Yes			Yes		Yes		Yes		
	60		Yes ⁵		Yes			Yes		Yes		Yes		
n47 ¹⁰	15		Yes		Yes		Yes	Yes						
	30		Yes		Yes		Yes	Yes						
	60		Yes		Yes		Yes	Yes						
n48	15	Yes ⁵	Yes	Yes	Yes			Yes	Yes ⁶					
	30		Yes	Yes	Yes			Yes	Yes ⁶	Yes ⁶		Yes ⁶	Yes ^{6,4}	Yes ⁶
	60		Yes	Yes	Yes			Yes	Yes ⁶	Yes ⁶		Yes ⁶	Yes ^{6,4}	Yes ⁶
n50	15	Yes ⁵	Yes	Yes	Yes		Yes	Yes	Yes					
	30		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes ³		
	60		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes ³		
n51	15	Yes												
	30													
	60													
n53	15	Yes	Yes											
	30		Yes											
	60		Yes											
n65	15	Yes	Yes	Yes	Yes				Yes					
	30		Yes	Yes	Yes				Yes					
	60		Yes	Yes	Yes				Yes					
n66	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes						
	30		Yes	Yes	Yes	Yes	Yes	Yes						
	60		Yes	Yes	Yes	Yes	Yes	Yes						
n70	15	Yes	Yes	Yes	Yes ³	Yes ³								
	30		Yes	Yes	Yes ³	Yes ³								
	60		Yes	Yes	Yes ³	Yes ³								
n71	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									

	60													
n74	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60		Yes	Yes	Yes									
n75	15	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
n76	15	Yes												
	30													
	60													
n77	15		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes ⁴	Yes	Yes ⁴	Yes
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes ⁴	Yes	Yes ⁴	Yes
n78	15		Yes	Yes	Yes	Yes	Yes	Yes	Yes					
	30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes ⁴	Yes	Yes	Yes
	60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes ⁴	Yes	Yes	Yes
n79	15							Yes	Yes					
	30							Yes	Yes	Yes		Yes		Yes
	60							Yes	Yes	Yes		Yes		Yes
n80	15	Yes	Yes	Yes	Yes	Yes	Yes							
	30		Yes	Yes	Yes	Yes	Yes							
	60		Yes	Yes	Yes	Yes	Yes							
n81	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n82	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n83	15	Yes	Yes	Yes	Yes ⁷									
	30		Yes	Yes	Yes ⁷									
	60													
n84	15	Yes	Yes	Yes	Yes									

	30		Yes	Yes	Yes									
	60		Yes	Yes	Yes									
n86	15	Yes	Yes	Yes	Yes			Yes						
	30		Yes	Yes	Yes			Yes						
	60		Yes	Yes	Yes			Yes						
n89	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n90	15		Yes	Yes	Yes		Yes	Yes	Yes					
	30		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes	Yes	Yes
	60		Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes	Yes	Yes
n91	15	Yes	Yes ⁸											
	30													
	60													
n92	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n93	15	Yes	Yes ⁸											
	30													
	60													
n94	15	Yes	Yes	Yes	Yes									
	30		Yes	Yes	Yes									
	60													
n95	15	Yes	Yes	Yes										
	30		Yes	Yes										
	60		Yes	Yes										
n96	15				Yes			Yes						
	30				Yes			Yes		Yes		Yes		
	60				Yes			Yes		Yes		Yes		
NOTE 1: Void. NOTE 2: Void. NOTE 3: This UE channel bandwidth is applicable only to downlink. NOTE 4: This UE channel bandwidth is optional in this release of the specification.														

NOTE 5:	For this bandwidth, the minimum requirements are restricted to operation when carrier is configured as an SCell part of DC or CA configuration.
NOTE 6:	For this bandwidth, the minimum requirements are restricted to operation when carrier is configured as a downlink SCell part of CA configuration.
NOTE 7:	For UEs supporting up to 30 MHz channel bandwidth, the minimum requirements are specified for any NR UL channel bandwidth confined to 703-733 MHz or 718-748 MHz.
NOTE 8:	This UE channel bandwidth is applicable only to uplink.
NOTE 9:	Void.
NOTE 10:	For this band, UE channel bandwidths which are applicable to sidelink operation are specified in Table 5.3E.1-1.

5.3.6 Asymmetric channel bandwidths

The UE channel bandwidth can be asymmetric in downlink and uplink. In asymmetric channel bandwidth operation, the narrower carrier shall be confined within the frequency range of the wider channel bandwidth.

In FDD, the confinement is defined as a deviation to the Tx-Rx carrier center frequency separation (defined in table 5.4.4-1) as following:

$$\Delta F_{\text{TX-RX}} = | (BW_{\text{DL}} - BW_{\text{UL}}) / 2 |$$

The operating bands and supported asymmetric channel bandwidth combinations are defined in table 5.3.6-1.

Table 5.3.6-1: FDD asymmetric UL and DL channel bandwidth combinations

NR Band	Channel bandwidths for UL (MHz)	Channel bandwidths for DL (MHz)	Asymmetric channel bandwidth combination set
n66	5, 10	20, 40	0
	20	40	
	5, 10	20, 25, 30, 40	1
	20, 25, 30	40	
n70	5, 10	15	0
	5, 10, 15	20, 25	
n71	5	10	0
	10	15	
	15	20	
n91 ¹	10	5	0
n92 ¹	5	10, 15, 20	0
	10	15, 20	
n93 ¹	10	5	0

n94 ¹	5	10, 15, 20	0
	10	15, 20	
NOTE 1: The assignment of the paired UL and DL channels are subject to a TX-RX separation as specified in clause 5.4.4.			

In TDD, the operating bands and supported asymmetric channel bandwidth combinations are defined in table 5.3.6-2.

Table 5.3.6-2: TDD asymmetric UL and DL channel bandwidth combinations

NR Band	Channel bandwidths for UL (MHz)	Channel bandwidths for DL (MHz)	Asymmetric channel bandwidth combination set
n50	60	80	0
NOTE 1: Both centre frequency and BWP-ID shall match between DL and UL carriers as defined in TS 38.331 [7] cl. 6.3.2 and TS 38.213 [8] clause 12.			
NOTE 2: In a case a UE is configured with a full width of BWP within both UL/ DL channels, the centre frequency of UL/ DL channels shall be same.			
NOTE 3: A position of Point A is common between UL and DL carriers as defined in TS 38.331 [7] cl. 6.3.2.			
NOTE 4: The maximum transmission bandwidth configuration N_{RB} for 60 MHz UL channel BW is reduced from the value defined in table 5.3.2-1 to 161 for 30kHz SCS for Asymmetric channel bandwidth combination set 0 for n50.			

5.3A UE channel bandwidth for CA

5.3A.1 General

Figure 5.3A.1-1: Void

Figure 5.3A.1-2: Void

5.3A.2 Maximum transmission bandwidth configuration for CA

For carrier aggregation, the maximum transmission bandwidth configuration is defined per component carrier and the requirement is specified in clause 5.3.2.

5.3A.3 Minimum guardband and transmission bandwidth configuration for CA

Resource block

The upper sub-block edge of the Sub-block Bandwidth is defined as

$$F_{\text{edge,block,high}} = F_{\text{C,block,high}} + F_{\text{offset,high}}.$$

The Sub-block Bandwidth, $BW_{\text{Channel,block}}$, is defined as follows:

$$BW_{\text{Channel,block}} = F_{\text{edge,block,high}} - F_{\text{edge,block,low}} \text{ (MHz)}$$

The lower and upper frequency offsets $F_{\text{offset,block,low}}$ and $F_{\text{offset,block,high}}$ depend on the transmission bandwidth configurations of the lowest and highest assigned edge component carriers within a sub-block and are defined as

$$F_{\text{offset,block,low}} = (N_{\text{RB,low}} * 12 + 1) * SCS_{\text{low}} / 2 + BW_{\text{GB}} \text{ (MHz)}$$

$$F_{\text{offset,block,high}} = (N_{\text{RB,high}} * 12 - 1) * SCS_{\text{high}} / 2 + BW_{\text{GB}} \text{ (MHz)}$$

$$BW_{\text{GB}} = \max(BW_{\text{GB,Channel}(k)})$$

where $N_{\text{RB,low}}$ and $N_{\text{RB,high}}$ are the transmission bandwidth configurations according to Table 5.3.2-1 for the lowest and highest assigned component carrier within a sub-block, respectively. SCS_{low} and SCS_{high} are the sub-carrier spacing for the lowest and highest assigned component carrier within a sub-block, respectively. SCS_{low} , SCS_{high} , $N_{\text{RB,low}}$, $N_{\text{RB,high}}$, and $BW_{\text{GB,Channel}(k)}$ use the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $BW_{\text{GB,Channel}(k)}$ is the minimum guard band for carrier k according to Table 5.3.3-1 for the said μ value. In case there is no common μ value for both of the channel bandwidths, $\mu=1$ is used for SCS_{low} , SCS_{high} , $N_{\text{RB,low}}$, $N_{\text{RB,high}}$, and $BW_{\text{GB,Channel}(k)}$.

The sub-block gap size between two consecutive sub-blocks W_{gap} is defined as

$$W_{\text{gap}} = F_{\text{edge,block } n+1,\text{low}} - F_{\text{edge,block } n,\text{high}} \text{ (MHz)}$$

5.3A.4 Void

5.3A.5 UE channel bandwidth per operating band for CA

The requirements for carrier aggregation in this specification are defined for carrier aggregation configurations.

For intra-band contiguous carrier aggregation, a carrier aggregation configuration is a single operating band supporting a carrier aggregation bandwidth class with associated bandwidth combination sets specified in clause 5.5A.1. For each carrier aggregation configuration, requirements are specified for all aggregated channel bandwidths contained in a bandwidth combination set, a UE can indicate support of several bandwidth combination sets per carrier aggregation configuration. For intra-band non-contiguous carrier aggregation, a carrier aggregation configuration is a single operating band supporting two or more sub-blocks, each supporting a carrier aggregation bandwidth class.

For intra-band non-contiguous uplink carrier aggregation, frequency separation class (Fs) specified in Table 5.3A.5-2 indicates the maximum frequency span between lower edge of lowest component carrier and upper edge of highest component carrier that UE can support per band combination in uplink in non-contiguous intra-band operation when the signalling is absent for dualPA-Architecture IE.

For inter-band carrier aggregation, a carrier aggregation configuration is a combination of operating bands, each supporting a carrier aggregation bandwidth class.

Table 5.3A.5-1: NR CA bandwidth classes

NR CA bandwidth class	Aggregated channel bandwidth	Number of contiguous CC	Fallback group
A	$BW_{\text{Channel}} \leq BW_{\text{Channel,max}}$	1	1, 2, 3 ⁴
B	$20 \text{ MHz} \leq BW_{\text{Channel_CA}} \leq 100 \text{ MHz}$	2	2, 3 ⁴
C	$100 \text{ MHz} < BW_{\text{Channel_CA}} \leq 2 \times BW_{\text{Channel,max}}$	2	1, 3 ⁴
D	$200 \text{ MHz} < BW_{\text{Channel_CA}} \leq 3 \times BW_{\text{Channel,max}}$	3	
E	$300 \text{ MHz} < BW_{\text{Channel_CA}} \leq 4 \times BW_{\text{Channel,max}}$	4	
G	$100 \text{ MHz} < BW_{\text{Channel_CA}} \leq 150 \text{ MHz}$	3	2
H	$150 \text{ MHz} < BW_{\text{Channel_CA}} \leq 200 \text{ MHz}$	4	
I	$200 \text{ MHz} < BW_{\text{Channel_CA}} \leq 250 \text{ MHz}$	5	
J	$250 \text{ MHz} < BW_{\text{Channel_CA}} \leq 300 \text{ MHz}$	6	
K	$300 \text{ MHz} < BW_{\text{Channel_CA}} \leq 350 \text{ MHz}$	7	
L	$350 \text{ MHz} < BW_{\text{Channel_CA}} \leq 400 \text{ MHz}$	8	
M ³	$50 \text{ MHz} \leq BW_{\text{Channel_CA}} \leq 200 \text{ MHz}$	3	3 ⁴
N ³	$80 \text{ MHz} \leq BW_{\text{Channel_CA}} \leq 300 \text{ MHz}$	4	
O ³	$100 \text{ MHz} \leq BW_{\text{Channel_CA}} \leq 400 \text{ MHz}$	5	
NOTE 1: $BW_{\text{Channel,max}}$ is maximum channel bandwidth supported among all bands in a release NOTE 2: It is mandatory for a UE to be able to fallback to lower order NR CA bandwidth class configuration within a fallback group. It is not mandatory for a UE to be able to fallback to lower order NR CA bandwidth class configuration that belong to a different fallback group. NOTE 3: This bandwidth class is only applicable to bands identified for use with shared spectrum channel access in Table 5.2-1. NOTE 4: Fallback group 3 is only applicable to bands identified for use with shared spectrum channel access in Table 5.2-1.			

Table 5.3A.5-2: NR intra-band non-contiguous UL CA frequency separation classes

NR NC UL CA frequency separation class	Maximum allowed frequency separation
I	100 MHz
II	200 MHz
III	[600 MHz]

5.3E Channel bandwidth for V2X

5.3E.1 General

NR V2X operation channel bandwidths for each operating band is specified in Table 5.3E.1-1. The same (symmetrical) channel bandwidth is specified for both the transmission and reception path.

Table 5.3E.1-1 NR V2X operation channel bandwidths for each operating band

NR band / SCS / UE Channel bandwidth					
NR Band	SCS kHz	10 MHz	20 MHz	30 MHz	40 MHz
n38	15	Yes	Yes	Yes	Yes
	30	Yes	Yes	Yes	Yes
	60	Yes	Yes	Yes	Yes
n47	15	Yes	Yes	Yes	Yes
	30	Yes	Yes	Yes	Yes
	60	Yes	Yes	Yes	Yes

5.3E.2 Channel bandwidth for V2X concurrent operation

For NR V2X inter-band concurrent operation in FR1, the NR V2X channel bandwidths for each operating band is specified in Table 5.3E.2-1.

Table 5.3E.2-1: Inter-band concurrent operation configurations

V2X concurrent operating band Configuration	NR Bands	SCS kHz	5 MHz	10 MHz	15 MHz	20 MHz	30 MHz	40 MHz	50 MHz	Maximum bandwidth [MHz]	Bandwidth combination set
---	----------	---------	-------	--------	--------	--------	--------	--------	--------	-------------------------	---------------------------

V2X_n71A-n47A	n71	15	Yes	Yes	Yes	Yes				60	0
		30		Yes	Yes	Yes					
		60									
	n47	15		Yes		Yes	Yes	Yes			
		30		Yes		Yes	Yes	Yes			
		60		Yes		Yes	Yes	Yes			

5.4 Channel arrangement

5.4.1 Channel spacing

5.4.1.1 Channel spacing for adjacent NR carriers

The spacing between carriers will depend on the deployment scenario, the size of the frequency block available and the channel bandwidths. The nominal channel spacing between two adjacent NR carriers is defined as following:

- For NR operating bands with 100 kHz channel raster,

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2$$

- For NR operating bands with 15 kHz channel raster,

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2 + \{-5 \text{ kHz}, 0 \text{ kHz}, 5 \text{ kHz}\} \text{ for } \Delta F_{\text{Raster}} \text{ equals } 15 \text{ kHz}$$

$$\text{Nominal Channel spacing} = (BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)})/2 + \{-10 \text{ kHz}, 0 \text{ kHz}, 10 \text{ kHz}\} \text{ for } \Delta F_{\text{Raster}} \text{ equals } 30 \text{ kHz}$$

where $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are the channel bandwidths of the two respective NR carriers. The channel spacing can be adjusted depending on the channel raster to optimize performance in a particular deployment scenario.

For NR bands restricted to operation with shared-spectrum channel access, the maximum deviation from the nominal channel spacing is 40 kHz.

5.4.2 Channel raster

5.4.2.1 NR-ARFCN and channel raster

The global frequency channel raster defines a set of RF reference frequencies F_{REF} . The RF reference frequency is used in signalling to identify the position of RF channels, SS blocks and other elements.

The global frequency raster is defined for all frequencies from 0 to 100 GHz. The granularity of the global frequency raster is ΔF_{Global} .

RF reference frequencies are designated by an NR Absolute Radio Frequency Channel Number (NR-ARFCN) in the range (0... 2016666) on the global frequency raster. The relation between the NR-ARFCN and the RF reference frequency F_{REF} in MHz is given by the following equation, where $F_{\text{REF-Offs}}$ and $N_{\text{Ref-Offs}}$ are given in table 5.4.2.1-1 and N_{REF} is the NR-ARFCN.

$$F_{\text{REF}} = F_{\text{REF-Offs}} + \Delta F_{\text{Global}} (N_{\text{REF}} - N_{\text{REF-Offs}})$$

Table 5.4.2.1-1: NR-ARFCN parameters for the global frequency raster

Frequency range (MHz)	ΔF_{Global} (kHz)	$F_{\text{REF-Offs}}$ (MHz)	$N_{\text{REF-Offs}}$	Range of N_{REF}
0 – 3000	5	0	0	0 – 599999
3000 – 24250	15	3000	600000	600000 – 2016666

The channel raster defines a subset of RF reference frequencies that can be used to identify the RF channel position in the uplink and downlink. The RF reference frequency for an RF channel maps to a resource element on the carrier. For each operating band, a subset of frequencies from the global frequency raster are applicable for that band and forms a channel raster with a granularity ΔF_{Raster} , which may be equal to or larger than ΔF_{Global} .

For SUL bands except n95, for the uplink of all FDD bands defined in Table 5.2-1, and for TDD bands n34, n39, n48, n90 and n38,

$$F_{\text{REF, shift}} = F_{\text{REF}} + \Delta_{\text{shift}}, \Delta_{\text{shift}} = 0 \text{ kHz or } 7.5 \text{ kHz.}$$

where Δ_{shift} is signalled by the network in higher layer parameter *frequencyShift7p5khz* [7]. For Band n34, n38, n39 and n48 $F_{\text{REF, shift}}$ is only applicable to uplink transmissions using a 15 kHz SCS.

The mapping between the channel raster and corresponding resource element is given in Clause 5.4.2.2. The applicable entries for each operating band are defined in Clause 5.4.2.3.

5.4.2.2 Channel raster to resource element mapping

The mapping between the RF reference frequency on the channel raster and the corresponding resource element is given in Table 5.4.2.2-1 and can be used to identify the RF channel position. The mapping depends on the total number of RBs that are

allocated in the channel and applies to both UL and DL. The mapping must apply to at least one numerology supported by the UE.

Table 5.4.2.2-1: Channel raster to resource element mapping

	$N_{RB} \bmod 2 = 0$	$N_{RB} \bmod 2 = 1$
Resource element index	0	6
Physical resource block index		

N_{RB} is the maximum transmission bandwidth configuration specified in sub-clause 5.3.2, n_{PRB} is the PRB index within the N_{RB} , and i is the resource element index within this PRB.

5.4.2.3 Channel raster entries for each operating band

The RF channel positions on the channel raster in each NR operating band are given through the applicable NR-ARFCN in Table 5.4.2.31, using the channel raster to resource element mapping in clause 5.4.2.2.

For NR operating bands with 100 kHz channel raster, $\Delta F_{\text{Raster}} = 20 \times \Delta F_{\text{Global}}$. In this case every 20th NR-ARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 5.4.2.31 is given as <20>.

For NR operating bands with 15 kHz channel raster below 3GHz, $\Delta F_{\text{Raster}} = I \times \Delta F_{\text{Global}}$, where $I \in \{3,6\}$. Every I^{th} NR-ARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 5.4.2.31 is given as <I>.

For NR operating bands with 15 kHz channel raster above 3GHz, $\Delta F_{\text{Raster}} = I \times \Delta F_{\text{Global}}$, where $I \in \{1,2\}$. Every I^{th} NR-ARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in table 5.4.2.3-1 is given as <I>.

In frequency bands with two or more ΔF_{Raster} : For 15 kHz and 30 kHz channel raster the higher ΔF_{Raster} applies to channels using only the SCS that is equal to or larger than the higher ΔF_{Raster} and SSB SCS is equal to the higher ΔF_{Raster} .

Table 5.4.2.3-1: Applicable NR-ARFCN per operating band

NR operating band	ΔF_{Raster} (kHz)	Uplink Range of N_{REF} (First - <Step size> - Last)	Downlink Range of N_{REF} (First - <Step size> - Last)
n1	100	384000 - <20> - 396000	422000 - <20> - 434000
n2	100	370000 - <20> - 382000	386000 - <20> - 398000
n3	100	342000 - <20> - 357000	361000 - <20> - 376000
n5	100	164800 - <20> - 169800	173800 - <20> - 178800
n7	100	500000 - <20> - 514000	524000 - <20> - 538000

n8	100	176000 - <20> - 183000	185000 - <20> - 192000
n12	100	139800 - <20> - 143200	145800 - <20> - 149200
n14	100	157600 - <20> - 159600	151600 - <20> - 153600
n18	100	163000 - <20> - 166000	172000 - <20> - 175000
n20	100	166400 - <20> - 172400	158200 - <20> - 164200
n25	100	370000 - <20> - 383000	386000 - <20> - 399000
n26	100	162800 - <20> - 169800	171800 - <20> - 178800
n28	100	140600 - <20> - 149600	151600 - <20> - 160600
n29	100	N/A	143400 - <20> - 145600
n30	100	461000 - <20> - 463000	470000 - <20> - 472000
n34	100	402000 - <20> - 405000	402000 - <20> - 405000
n38	100	514000 - <20> - 524000	514000 - <20> - 524000
n39	100	376000 - <20> - 384000	376000 - <20> - 384000
n40	100	460000 - <20> - 480000	460000 - <20> - 480000
n41	15	499200 - <3> - 537999	499200 - <3> - 537999
	30	499200 - <6> - 537996	499200 - <6> - 537996
n46 ²	15	743334 - <1> - 795000	743334 - <1> - 795000
n47	15	790334 - <1> - 795000	790334 - <1> - 795000
n48	15	636667 - <1> - 646666	636667 - <1> - 646666
	30	636668 - <2> - 646666	636668 - <2> - 646666
n50	100	286400 - <20> - 303400	286400 - <20> - 303400
n51	100	285400 - <20> - 286400	285400 - <20> - 286400
n53	100	496700 - <20> - 499000	496700 - <20> - 499000
n65	100	384000 - <20> - 402000	422000 - <20> - 440000
n66	100	342000 - <20> - 356000	422000 - <20> - 440000
n70	100	339000 - <20> - 342000	399000 - <20> - 404000
n71	100	132600 - <20> - 139600	123400 - <20> - 130400
n74	100	285400 - <20> - 294000	295000 - <20> - 303600
n75	100	N/A	286400 - <20> - 303400
n76	100	N/A	285400 - <20> - 286400
n77	15	620000 - <1> - 680000	620000 - <1> - 680000
	30	620000 - <2> - 680000	620000 - <2> - 680000
n78	15	620000 - <1> - 653333	620000 - <1> - 653333

	30	620000 - <2> - 653332	620000 - <2> - 653332
n79	15	693334 - <1> - 733333	693334 - <1> - 733333
	30	693334 - <2> - 733332	693334 - <2> - 733332
n80	100	342000 - <20> - 357000	N/A
n81	100	176000 - <20> - 183000	N/A
n82	100	166400 - <20> - 172400	N/A
n83	100	140600 - <20> - 149600	N/A
n84	100	384000 - <20> - 396000	N/A
n86	100	342000 - <20> - 356000	N/A
n89	100	164800 - <20> - 169800	N/A
n90	15	499200 - <3> - 537999	499200 - <3> - 537999
	30	499200 - <6> - 537996	499200 - <6> - 537996
	100	499200 - <20> - 538000	499200 - <20> - 538000
n91	100	166400 - <20> - 172400	285400 - <20> - 286400
n92	100	166400 - <20> - 172400	286400 - <20> - 303400
n93	100	176000 - <20> - 183000	285400 - <20> - 286400
n94	100	176000 - <20> - 183000	286400 - <20> - 303400
n95	100	402000 - <20> - 405000	N/A
n96 ³	15	795000 - <1> - 875000	795000 - <1> - 875000
NOTE 1: The channel numbers that designate carrier frequencies so close to the operating band edges that the carrier extends beyond the operating band edge shall not be used.			
NOTE 2: The following N _{REF} are allowed for operation in Band n46: see Table 5.4.2.3-2.			
NOTE 3: The following N _{REF} are allowed for operation in Band n96: see Table 5.4.2.3-3.			

Table 5.4.2.3-2: Allowed N_{REF} (NR-ARFCN) for operation in Band n46

Channel Bandwidth	Allowed N _{REF}
10 MHz	782000, 788668
20 MHz	744000, 745332, 746668, 748000, 749332, 750668, 752000, 753332, 754668, 756000, 765332, 766668, 768000, 769332, 770668, 772000, 773332, 774668, 776000, 777332, 778668, 780000, 781332, 783000, 784332, 785668, 787000, 788332, 789668, 791000, 792332, 793668

40 MHz	744668, 746000, 748668, 751332, 754000, 755332, 766000, 767332, 770000, 772668, 775332, 778000, 780668, 783668, 786332, 787668, 790332, 793000
60 MHz	745332, 746668, 748000, 752000, 753332, 754668, 766668, 768000, 769332, 773332, 774668, 778668, 780000, 784332, 785668, 791000, 792332
80 MHz	746000, 747332, 752668, 754000, 767332, 768668, 774000, 779332, 785000, 791668
NOTE: 10 MHz channel bandwidth shall only apply in certain regions where the absence of non 3GPP technologies can be guaranteed on a long-term basis in this version of specification.	

Table 5.4.2.3-3: Allowed N_{REF} (NR-ARFCN) for operation in Band n96

Channel Bandwidth	Allowed N_{REF}
20 MHz	797000, 798332, 799668, 801000, 802332, 803668, 805000, 806332, 807668, 809000, 810332, 811668, 813000, 814332, 815668, 817000, 818332, 819668, 821000, 822332, 823668, 825000, 826332, 827668, 829000, 830332, 831668, 833000, 834332, 835668, 837000, 838332, 839668, 841000, 842332, 843668, 845000, 846332, 847668, 849000, 850332, 851668, 853000, 854332, 855668, 857000, 858332, 859668, 861000, 862332, 863668, 865000, 866332, 867668, 869000, 870332, 871668, 873000, 874332
40 MHz	797668, 800332, 803000, 805668, 808332, 811000, 813668, 816332, 819000, 821668, 824332, 827000, 829668, 832332, 835000, 837668, 840332, 843000, 845668, 848332, 851000, 853668, 856332, 859000, 861668, 864332, 867000, 869668, 872332
60 MHz	798332, 799668, 803668, 805000, 809000, 810332, 814332, 815668, 819668, 821000, 825000, 826332, 830332, 831668, 835668, 837000, 841000, 842332, 846332, 847668, 851668, 853000, 857000, 858332, 862332, 863668, 867668, 869000, 873000
80 MHz	799000, 804332, 809668, 815000, 820332, 825668, 831000, 836332, 841668, 847000, 852332, 857668, 863000, 868332

5.4.3 Synchronization raster

5.4.3.1 Synchronization raster and numbering

The synchronization raster indicates the frequency positions of the synchronization block that can be used by the UE for system acquisition when explicit signalling of the synchronization block position is not present.

A global synchronization raster is defined for all frequencies. The frequency position of the SS block is defined as SS_{REF} with corresponding number GSCN. The parameters defining the SS_{REF} and GSCN for all the frequency ranges are in Table 5.4.3.1-1. The resource element corresponding to the SS block reference frequency SS_{REF} is given in clause 5.4.3.2. The synchronization raster and the subcarrier spacing of the synchronization block is defined separately for each band.

Table 5.4.3.1-1: GSCN parameters for the global frequency raster

Frequency range	SS Block frequency position SS_{REF}	GSCN	Range of GSCN
0 – 3000 MHz	$N * 1200\text{kHz} + M * 50\text{ kHz}$, $N=1:2499$, $M \in \{1,3,5\}$ (Note 1)	$3N + (M-3)/2$	2 – 7498
3000 – 24250 MHz	$3000\text{ MHz} + N * 1.44\text{ MHz}$ $N = 0:14756$	$7499 + N$	7499 – 22255

NOTE 1: The default value for operating bands with which only support SCS spaced channel raster(s) is $M=3$.

5.4.3.2 Synchronization raster to synchronization block resource element mapping

The mapping between the synchronization raster and the corresponding resource element of the SS block is given in Table 5.4.3.2-1.

Table 5.4.3.2-1: Synchronization raster to SS block resource element mapping

Resource element index	120

is the subcarrier number of SS/PBCH block defined in TS 38.211 clause 7.4.3.1 [6].

5.4.3.3 Synchronization raster entries for each operating band

The synchronization raster for each band is give in Table 5.4.3.3-1. The distance between applicable GSCN entries is given by the <Step size> indicated in Table 5.4.3.3-1.

Table 5.4.3.3-1: Applicable SS raster entries per operating band

NR operating band	SS Block SCS	SS Block pattern ¹	Range of GSCN
-------------------	--------------	-------------------------------	---------------

			(First - <Step size> - Last)
n1	15 kHz	Case A	5279 – <1> – 5419
n2	15 kHz	Case A	4829 – <1> – 4969
n3	15 kHz	Case A	4517 – <1> – 4693
n5	15 kHz	Case A	2177 – <1> – 2230
	30 kHz	Case B	2183 – <1> – 2224
n7	15 kHz	Case A	6554 – <1> – 6718
n8	15 kHz	Case A	2318 – <1> – 2395
n12	15 kHz	Case A	1828 – <1> – 1858
n14	15 kHz	Case A	1901 – <1> – 1915
n18	15 kHz	Case A	2156 – <1> – 2182
n20	15 kHz	Case A	1982 – <1> – 2047
n25	15 kHz	Case A	4829 – <1> – 4981
n26	15 kHz	Case A	2153 – <1> – 2230
n28	15 kHz	Case A	1901 – <1> – 2002
n29	15 kHz	Case A	1798 – <1> – 1813
n30	15 kHz	Case A	5879 – <1> – 5893
n34	15 kHz	Case A	NOTE 5
	30 kHz	Case C	5036 – <1> – 5050
n38	15 kHz	Case A	NOTE 2
	30 kHz	Case C	6437 – <1> – 6538
n39	15 kHz	Case A	NOTE 6
	30 kHz	Case C	4712 – <1> – 4789
n40	30 kHz	Case C	5762 – <1> – 5989
n41	15 kHz	Case A	6246 – <3> – 6717
	30 kHz	Case C	6252 – <3> – 6714
n46 ³	30 kHz	Case C	8993 – <1> – 9530
n48	30 kHz	Case C	7884 – <1> – 7982
n50	30 kHz	Case C	3590 – <1> – 3781
n51	15 kHz	Case A	3572 – <1> – 3574
n53	15 kHz	Case A	6215 – <1> – 6232
	30 kHz	Case C	6221 – <1> – 6226
n65	15 kHz	Case A	5279 – <1> – 5494
n66	15 kHz	Case A	5279 – <1> – 5494
	30 kHz	Case B	5285 – <1> – 5488
n70	15 kHz	Case A	4993 – <1> – 5044
n71	15 kHz	Case A	1547 – <1> – 1624

n74	15 kHz	Case A	3692 – <1> – 3790
n75	15 kHz	Case A	3584 – <1> – 3787
n76	15 kHz	Case A	3572 – <1> – 3574
n77	30 kHz	Case C	7711 – <1> – 8329
n78	30 kHz	Case C	7711 – <1> – 8051
n79	30 kHz	Case C	8480 – <16> – 8880
n90	15 kHz	Case A	6246 – <1> – 6717
	30 kHz	Case C	6252 – <1> – 6714
n91	15 kHz	Case A	3572 – <1> – 3574
n92	15 kHz	Case A	3584 – <1> – 3787
n93	15 kHz	Case A	3572 – <1> – 3574
n94	15 kHz	Case A	3584 – <1> – 3787
n96 ⁴	30 kHz	Case C	9531 – <1> – 10363

NOTE 1: SS Block pattern is defined in clause 4.1 in TS 38.213 [8].
NOTE 2: The applicable SS raster entries are GSCN = {6432, 6443, 6457, 6468, 6479, 6493, 6507, 6518, 6532, 6543}.
NOTE 3: The following GSCN are allowed for operation in band n46:
GSCN = {8996, 9010, 9024, 9038, 9051, 9065, 9079, 9093, 9107, 9121, 9218, 9232, 9246, 9260, 9274, 9288, 9301, 9315, 9329, 9343, 9357, 9371, 9385, 9402, 9416, 9430, 9444, 9458, 9472, 9485, 9499, 9513}.
NOTE 4: The following GSCN are allowed for operation in band n96:
GSCN = {9548, 9562, 9576, 9590, 9603, 9617, 9631, 9645, 9659, 9673, 9687, 9701, 9714, 9728, 9742, 9756, 9770, 9784, 9798, 9812, 9826, 9840, 9853, 9867, 9881, 9895, 9909, 9923, 9937, 9951, 9964, 9978, 9992, 10006, 10020, 10034, 10048, 10062, 10076, 10090, 10103, 10117, 10131, 10145, 10159, 10173, 10187, 10201, 10214, 10228, 10242, 10256, 10270, 10284, 10298, 10312, 10325, 10339, 10353}.
NOTE 5: The applicable SS raster entries are GSCN = {5032, 5043, 5054}
NOTE 6: The applicable SS raster entries are GSCN = {4707, 4715, 4718, 4729, 4732, 4743, 4747, 4754, 4761, 4768, 4772, 4782, 4786, 4793}

5.4.4 TX–RX frequency separation

The default TX channel (carrier centre frequency) to RX channel (carrier centre frequency) separation for operating bands is specified in Table 5.4.4-1.

Table 5.4.4-1: UE TX-RX frequency separation

NR Operating Band	TX – RX carrier centre frequency separation
n1	190 MHz
n2	80 MHz
n3	95 MHz

n5	45 MHz
n7	120 MHz
n8	45 MHz
n12	30 MHz
n14	-30 MHz
n18	45 MHz
n20	-41 MHz
n25	80 MHz
n26	45 MHz
n28	55 MHz
n30	45 MHz
n65	190 MHz
n66	400 MHz
n70	300 MHz
n71	-46 MHz
n74	48 MHz
n91	570 MHz – 595 MHz (NOTE 2)
n92	575 MHz – 680 MHz ($\mu = 0$) 580 MHz – 675 MHz ($\mu = 1$) (NOTE 2)
n93	517 MHz – 547 MHz (NOTE 2)
n94	522 MHz – 632 MHz ($\mu = 0$) 527 MHz – 627 MHz ($\mu = 1$) (NOTE 2)
NOTE 1: Void	
NOTE 2: The range of TX-RX frequency separation given paired UL and DL channel bandwidths BW_{UL} and BW_{DL} is given by the respective lower and upper limit $F_{DL_low} - F_{UL_high} + 0.5(BW_{DL} + BW_{UL})$ and $F_{DL_high} - F_{UL_low} - 0.5(BW_{DL} + BW_{UL})$. The UL and DL channel bandwidth combinations specified in Table 5.3.5-1 and 5.3.6-1 depend on the subcarrier spacing configuration μ [6].	

5.4A Channel arrangement for CA

5.4A.1 Channel spacing for CA

For intra-band contiguous carrier aggregation with two or more component carriers, the nominal channel spacing between two adjacent NR component carriers is defined as the following unless stated otherwise:

For NR operating bands with a 100 kHz channel raster:

while for NR operating bands without a 100 kHz channel raster:

with

$$n = \mu_0$$

where $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are the channel bandwidths of the two respective NR component carriers according to Table 5.3.2-1 with values in MHz, μ_0 is the largest μ value among the subcarrier spacing configurations supported in the operating band for both of the channel bandwidths according to Table 5.3.5-1 and $GB_{\text{Channel}(i)}$ is the minimum guard band for channel bandwidth i according to Table 5.3.3-1 for the said μ value with μ as defined in TS 38.211. In case there is no common μ value for both of the channel bandwidths, $\mu_0=1$ is selected and $GB_{\text{Channel}(i)}$ is the minimum guard band for channel bandwidth i according to Table 5.3.3-1 for $\mu=1$ with μ as defined in TS 38.211.

The channel spacing for intra-band contiguous carrier aggregation can be adjusted to any multiple of least common multiple of channel raster and sub-carrier spacing less than the nominal channel spacing to optimize performance in a particular deployment scenario.

For intra-band contiguous carrier aggregation in NR bands restricted to operation with shared-spectrum channel access, the maximum deviation from the nominal channel spacing is 300 kHz.

For intra-band non-contiguous carrier aggregation, the channel spacing between two NR component carriers in different sub-blocks shall be larger than the nominal channel spacing defined in this clause.

5.4A.2 Channel raster for CA

For inter-band and intra-band carrier aggregation, the channel raster requirements in clause 5.4.2 apply for each operating band.

5.4A.3 Synchronization raster for CA

For inter-band and intra-band carrier aggregation, the synchronization raster requirements in clause 5.4.3 apply for each operating band.

5.4A.4 Tx-Rx frequency separation for CA

For inter-band carrier aggregation, the Tx-Rx frequency separation requirements in clause 5.4.4 apply for each operating band.

For intra-band carrier aggregation, the same TX-RX frequency separation as specified in Table 5.4.4-1 is applied to PCC and SCC, respectively.

5.4B Reserved

5.4C Reserved

5.4D Reserved

5.4E Channel arrangement for V2X

5.4E.1 Channel spacing

For NR V2X, the channel spacing requirements in clause 5.4.1 apply for each operating band.

5.4E.2 Channel raster

5.4E.2.1 NR-ARFCN and channel raster

For NR V2X, the NR-ARFCN and channel raster requirements in clause 5.4.2.1 apply for each operating band.

For NR V2X UE, the reference frequency can be shifted by configuration.

$$F_{\text{REF_V2X}} = F_{\text{REF}} + \Delta_{\text{shift}} + N * 5 \text{ kHz}$$

where

$\Delta_{\text{shift}} = 0 \text{ kHz}$ or 7.5 kHz indicated in IE (*frequencyShift7p5khz*), and

N can be set as one of following values $\{-1, 0, 1\}$, which are signalled by the network in higher layer parameters or configured by pre-configuration parameters.

5.4E.2.2 Channel raster to resource element mapping

For NR V2X, the channel raster to resource element mapping requirements in clause 5.4.2.2 apply for each operating band.

5.4E.2.3 Channel raster entries for each operating band

For NR V2X, the channel raster entries, the channel raster entries requirements in clause 5.4.2.3 apply for each operating band.

The RF channel positions on the channel raster in each NR V2X operating band are given through the applicable NR-ARFCN in Table 5.4.2.3-1, using the channel raster to resource element mapping in clause 5.4E.2.2.

For NR V2X operating band n47, $\Delta F_{\text{Raster}} = I \times \Delta F_{\text{Global}}$, where $I \in \{0, 1, 2, 3, 4, 5\}$. Every I^{th} NRARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 5.4.2.3-1 is given as $\langle I \rangle$.

5.4E.3 Synchronization raster for V2X

There is no synchronization raster definition for NR V2X for both licensed bands and unlicensed bands.

5.5 Void

5.5A Configurations for CA

5.5A.0 General

The configurations for CA operating band including Band n41 also apply for the corresponding CA operating bands with Band n90 replacing Band n41 but with otherwise identical parameters. For brevity the said configuration for CA operating bands with Band n90 are not listed in the tables below but are covered by this specification.

Noncontiguous resource allocation and almost contiguous allocation are not applicable for each NR carrier of intraband contiguous and non-contiguous CA configurations.

For a CA configuration with one or more operating band supporting asymmetric channel bandwidths as specified in sub-clause 5.3.6, requirements are defined for an asymmetric UL and DL channel bandwidth combination of a supported asymmetric channel bandwidth combination set for an operating band of the CA configuration when the said UL and DL channel bandwidths are also contained in a supported bandwidth combination set of the CA configuration.

For a higher order band combination of which CA_n20-n28 is a subset, the frequency range in band n28 is restricted for the higher order band combination to 703-733 MHz for the UL and 758-788 MHz for the DL.

5.5A.1 Configurations for intra-band contiguous CA

Table 5.5A.1-1: NR CA configurations and bandwidth combination sets defined for intra-band contiguous CA

NR CA configuration / Bandwidth combination set								
NR CA configuration	Uplink CA configurations	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Maximum aggregated bandwidth (MHz)	Bandwidth combination set
CA_n1B	-	10	10,15				40	0
		15	15,20					
		20	20					
CA_n7B	CA_n7B	10	10, 15, 20, 30, 40				50	0
		15	15, 20, 30					

		20	20, 30					
CA_n40B	-	20	80				100	0
		50	50					
CA_n41B	CA_n41B	10, 20, 30, 40, 50	10, 20, 30, 40, 50				100	0
CA_n41C	CA_n41C	40	80, 100				180	0
		50, 60, 80	60, 80, 100					
		10	100				190	1
		15, 20	90, 100					
		40	80, 90, 100					
		50, 60, 80, 90	60, 80, 90, 100					
CA_n46B	-	20, 40, 60	20, 40				100	0
CA_n46C	-	60, 80	60, 80				160	0
CA_n46D	-	60, 80	80	80			240	0
CA_n46M	-	20, 40, 60	20, 40	20, 40			140	0
CA_n46N	-	Void						0
	-	20, 40, 60	20, 40	20, 40	20, 40		180	1
CA_n46O	-	20, 60	20, 40	20, 40	20, 40	20, 40	220	0
CA_n48B	CA_n48B	5	15, 20				40	0
		10, 15, 20	10, 15, 20					
		15, 20	15, 20					
	-	10	50, 60, 80, 90				100	1
		15, 20	40, 50, 60, 80					
		40	40, 50, 60					
CA_n48C	-	10	100				140	0
		15	90, 100					
		20	90, 100					
		40	80, 90, 100					
CA_n66B	-	5	20, 40				50	0
		10	15, 20, 40					
		15	15, 20					

CA_n71B	-	5	20				25	0
		10	15					
		10	20				35	1
		15	15, 20					
CA_n77C	CA_n77C	50	60, 80, 100				200	0
		60	60, 80, 100					
		80	80, 100					
		100	100					
		10	100				200	1
		15, 20	90, 100					
		25, 30	80, 90, 100					
		40	70, 80, 90, 100					
		50, 60, 70, 80, 90, 100	60, 70, 80, 90, 100					
CA_n77D	-	100	100	100			300	0
CA_n78B	-	20	50				70	0
CA_n78C	CA_n78C	50	60, 80, 100				200	0
		60	60, 80, 100					
		80	80, 100					
		100	100					
		10	100				200	1
		15, 20	90, 100					
		25, 30	80, 90, 100					
		40	70, 80, 90, 100					
		50, 60, 70, 80, 90, 100	60, 70, 80, 90, 100					
CA_n78D	-	100	100	100			300	0
CA_n79C	CA_n79C	50	60, 80, 100				200	0
		60	60, 80, 100					

		80	80, 100					
		100	100					
CA_n79D	-	100	100	100			300	0
NOTE 1: For each channel bandwidth of each component carrier, refer to Table 5.3.5-1 for the applicable SCSs. For a given band, not all UE channel bandwidths support the same SCSs.								

Table 5.5A.1-2: Void

5.5A.2 Configurations for intra-band non-contiguous CA

Table 5.5A.2-1: NR CA configurations and bandwidth combination sets defined for intra-band non-contiguous CA

NR CA Configuration	Uplink Configurations	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Channel bandwidths for carrier (MHz)	Maximum Aggregated bandwidth (MHz)	Bandwidth combination set
CA_n3(2A)	-	5, 10, 15, 20	5, 10, 15, 20			40	0
CA_n7(2A)	-	5, 10, 15, 20	5, 10, 15, 20			40	0
CA_n25(2A)	-	5, 10, 15, 20	5, 10, 15, 20			40	0
CA_n41(2A)	CA_n41(2A)	40, 50, 60, 80	40, 50, 60, 80, 100			180	0
		10, 15, 20, 40, 50, 60, 80, 90	10, 15, 20, 40, 50, 60, 80, 90, 100			190	1
CA_n48(2A)	-	10, 15, 20, 40, 50, 60	10, 15, 20, 40, 50, 60, 80, 90, 100			140 ²	0
CA_n48(3A)	-	10, 15, 20, 40, 50, 60, 80, 90, 100	10, 15, 20, 40, 50, 60, 80, 90, 100	10, 15, 20, 40, 50, 60, 80, 90, 100		140 ²	0
CA_n48(4A)	-	10, 15, 20, 40, 50, 60, 80, 90, 100	10, 15, 20, 40, 50, 60, 80, 90, 100	10, 15, 20, 40, 50, 60, 80, 90, 100	10, 15, 20, 40, 50, 60, 80, 90, 100	135 ²	0
CA_n66(2A)	-	5, 10, 15, 20	5, 10, 15, 20, 40			60	0

CA_n77(2A)	CA_n77(2A)	20, 40, 80, 100	20, 40, 80, 100			200	0
CA_n78(2A)	CA_n78(2A)	10, 20, 40, 50, 60, 80, 90, 100	10, 20, 40, 50, 60, 80, 90, 100			200	0
		10, 20, 25, 30, 40, 50, 60, 80, 90, 100	10, 20, 25, 30, 40, 50, 60, 80, 90, 100			200	1
		10, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100	10, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100			200	2
NOTE 1: Void.							
NOTE 2: Parameter value accounts for both, the maximum frequency range of band n48 (150 MHz), and the minimum frequency gaps in between NR non-contiguous component carriers.							
NOTE 3: For each channel bandwidth of each component carrier, refer to Table 5.3.5-1 for the applicable SCSs. For a given band, not all UE channel bandwidths support the same SCSs.							

5.5A.3 Configurations for inter-band CA

- Table 5.5A.3-1: Void
- Table 5.5A.3-2: Void
- Table 5.5A.3-3: Void

5.5A.3.1 Configurations for inter-band CA (two bands)

Table 5.5A.3.1-1: NR CA configurations and bandwidth combinations sets defined for inter-band CA (two bands)

[illegible]

		n48	5	10	15	20			40	50 ⁴	60 ⁴		80 ⁴	90 ⁴	100 ⁴	
CA_n2A-n48C	CA_n2A-n48A	n2	5	10	15	20										0
		n48	See CA_n48C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n2A-n66A	-	n2	5	10	15	20										0
		n66	5	10	15	20			40							
CA_n2A-n77A	CA_n2A-n77A	n2	5	10	15	20										0
		n77		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n2A-n78A	CA_n2A-n78A	n2	5	10	15	20										0
		n78		10	15	20	25	30	40	50	60		80	90	100	
CA_n2A-n78(2A)	CA_n2A-n78A	n2	5	10	15	20										0
		n78	See CA_n78(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1													
CA_n3A-n7A	CA_n3A-n7A	n3	5	10	15	20	25	30								0
		n7	5	10	15	20	25	30	40	50						
CA_n3A-n7B	-	n3	5	10	15	20	25	30								0
		n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n3A-n8A	CA_n3A-n8A	n3	5	10	15	20	25	30								0
		n8	5	10	15	20										
CA_n3A-n28A	CA_n3A-n28A	n3	5	10	15	20	25	30								0
		n28	5	10	15	20										
CA_n3A-n38A	CA_n3A-n38A	n3	5	10	15	20	25	30								0
		n38	5	10	15	20			40							
CA_n3A-n40A	CA_n3A-n40A	n3	5	10	15	20	25	30								0
		n40	5	10	15	20	25	30	40	50	60		80			
CA_n3A-n41A	CA_n3A-n41A	n3	5	10	15	20	25	30								0
		n41		10	15	20			40	50	60		80	90	100	
		n3	5	10	15	20	25	30								1
		n41		10	15	20			40	50	60					
CA_n3A-n41C	CA_n3A-n41A	n3	5	10	15	20	25	30								0
		n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n3A-n41(2A)	CA_n3A-n41A	n3	5	10	15	20	25	30								0
		n41	See CA_n41(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n3A-n77A	CA_n3A-n77A	n3	5	10	15	20	25	30								0
		n77		10	15	20			40	50	60		80	90	100	
CA_n3A-n77(2A)	CA_n3A-n77A	n3	5	10	15	20	25	30								0
		n77	See CA_n77(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n3A-n78A	CA_n3A-n78A	n3	5	10	15	20	25	30								0
		n78		10	15	20			40	50	60		80	90	100	

CA_n3A-n78C	CA_n3A-n78A	n3	5	10	15	20	25	30								0
		n78	See CA_n78C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n3A-n78(2A)	-	n3	5	10	15	20	25	30								0
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n3A-n79A	CA_n3A-n79A	n3	5	10	15	20	25	30								0
		n79							40	50	60		80		100	
CA_n3A-n79C	CA_n3A-n79A	n3	5	10	15	20	25	30								0
		n79	See CA_n79C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n5A-n7A	-	n5	5	10	15	20										0
		n7	5	10	15	20	25	30	40	50						
CA_n5A-n7B	-	n5	5	10	15	20										0
		n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n5A-n66A	CA_n5A-n66A	n5	5	10	15	20										0
		n66	5	10	15	20			40							
CA_n5A-n77A	CA_n5A-n77A	n5	5	10	15	20										0
		n77		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n5A-n78A	CA_n5A-n78A	n5	5	10	15	20										0
		n78		10	15	20			40	50	60		80	90	100	
CA_n5A-n78C	CA_n5A-n78A	n5	5	10	15	20										0
		n78	See CA_n78C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n5A-n79A	CA_n5A-n79A	n5	5	10	15	20										0
		n79							40	50	60		80		100	
CA_n5A-n79C	CA_n5A-n79A	n5	5	10	15	20										0
		n79	See CA_n79C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n7A-n25A	CA_n7A-n25A	n7	5	10	15	20	25	30	40							0
		n25	5	10	15	20	25	30	40							
CA_n7A-n25(2A)	CA_n7A-n25A	n7	5	10	15	20	25	30	40							0
		n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n7(2A)-n25A	CA_n7A-n25A	n25	5	10	15	20	25	30	40							0
		n7	See CA_n7(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n7(2A)-n25(2A)	CA_n7A-n25A	n7	See CA_n7(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n7A-n28A	CA_n7A-n28A	n7	5	10	15	20	25	30	40	50						0
		n28	5	10	15	20										
CA_n7B-n28A	-	n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													0

		n28	5	10	15	20										
CA_n7A-n66A	CA_n7A-n66A	n7	5	10	15	20										0
		n66		10	15	20			40							
CA_n7A-n78A	CA_n7A-n78A	n7	5	10	15	20										0
		n78		10	15	20			40	50	60		80	90	100	
CA_n7A-n78(2A)	CA_n7A-n78A	n7	5	10	15	20	25	30	40	50						0
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n7(2A)-n78A	CA_n7A-n78A	n7	See CA_n7(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n78		10	15	20			40	50	60		80	90	100	
CA_n7(2A)-n78(2A)	CA_n7A-n78A	n7	See CA_n7(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n8A-n39A	CA_n8A-n39A	n8	5	10	15	20										0
		n39	5	10	15	20	25	30	40							
CA_n8A-n40A	CA_n8A-n40A	n8	5	10	15	20										0
		n40	5	10	15	20	25	30	40	50	60		80			
CA_n8A-n41A	CA_n8A-n41A	n8	5	10	15	20										0
		n41		10	15	20			40	50	60		80	90	100	
		n8	5	10	15	20										1
		n41		10	15	20			40	50	60					
CA_n8A-n75A	-	n8	5	10	15	20										0
		n75	5	10	15	20										
CA_n8A-n78A	CA_n8A-n78A	n8	5	10	15	20										0
		n78		10	15	20			40	50	60		80	90	100	
CA_n8A-n79A	CA_n8A-n79A	n8	5	10	15	20										0
		n79							40	50	60		80		100	
CA_n20A-n28A	CA_n20A-n28A	n20	5	10	15	20										0
		n28	5	10	15	20										
CA_n20A-n75A	-	n20	5	10	15	20										0
		n75	5	10	15	20										
CA_n20A-n78A	CA_n20A-n78A	n20	5	10	15	20										0
		n78		10	15	20			40	50	60		80	90	100	
CA_n25A-n41A	CA_n25A-n41A	n25	5	10	15	20										0
		n41		10	15	20			40	50	60		80	90	100	
CA_n25(2A)-n41A	CA_n25A-n41A	n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n41		10	15	20			40	50	60		80	90	100	
CA_n25A-n41C	CA_n25A-n41A	n25	5	10	15	20										0
		n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n25A-n41(2A)	CA_n25A-n41A	n25	5	10	15	20										0

		n41	See CA_n41(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1													
CA_n25A-n66A	CA_n25A-n66A	n25	5	10	15	20	25	30	40							0
		n66	5	10	15	20		30	40							
CA_n25A-n66(2A)	CA_n25A-n66A	n25	5	10	15	20	25	30	40							0
		n66	See CA_n66(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n25(2A)-n66A	CA_n25A-n66A	n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n66		10	15	20		30	40							
CA_n25(2A)-n66(2A)	CA_n25A-n66A	n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n66	See CA_n66(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n25A-n71A	CA_n25A-n71A	n25	5	10	15	20										0
		n71	5	10	15	20										
CA_n25A-n78A	CA_n25A-n78A	n25	5	10	15	20	25	30	40							0
		n78		10	15	20	25	30	40	50	60		80	90	100	
CA_n25A-n78(2A)	CA_n25A-n78A	n25	5	10	15	20	25	30	40							0
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n25(2A)-n78A	CA_n25A-n78A	n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n78		10	15	20	25	30	40	50	60		80	90	100	
CA_n25(2A)-n78(2A)	CA_n25A-n78A	n25	See CA_n25(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n78	See CA_n78(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1													
CA_n25A-n46A	-	n25	5	10	15	20										0
		n46				20			40		60		80			
CA_n28A-n40A	CA_n28A-n40A	n28	5	10	15	20										0
		n40	5	10	15	20	25	30	40	50	60		80			
CA_n28A-n41A	CA_n28A-n41A	n28	5	10	15	20										0
		n41		10	15	20			40	50	60		80	90	100	
CA_n28A-n50A	CA_n28A-n50A	n28	5	10	15	20										0
		n50	5	10	15	20			40	50	60		80 ¹			
CA_n28A-n75A	-	n28	5	10	15	20										0
		n75	5	10	15	20										
CA_n28A-n75A	-	n28	5	10	15	20										1
		n75	5	10	15	20	25	30	40	50						
CA_n28A-n77A	CA_n28A-n77A	n28	5	10	15	20										0
		n77		10	15	20			40	50	60		80	90	100	

CA_n28A-n77(2A)	CA_n28A-n77A	n28	5	10	15	20										0
		n77	See CA_n77(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n28A-n78A	CA_n28A-n78A	n28	5	10	15	20										0
		n78		10	15	20			40	50	60		80	90	100	
CA_n28A-n78(2A)	CA_n28A-n78A	n28	5	10	15	20										0
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n29A-n66A	-	n29	5	10												0
		n66	5	10	15	20			40							
CA_n29A-n66B	-	n29	5	10												0
		n66	See CA_n66B Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n29A-n66(2A)	-	n29	5	10												0
		n66	See CA_n66(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n29A-n70A	-	n29	5	10												0
		n70	5	10	15	20 ¹	25 ¹									
CA_n38A-n66A	CA_n38A-n66A	n38	5	10	15	20										0
		n66	5	10	15	20		30	40							
CA_n38A-n78A	CA_n38A-n78A	n38	5	10	15	20										0
		n78		10	15	20	25	30	40	50	60		80	90	100	
CA_n38A-n78(2A)	CA_n38A-n78A	n38	5	10	15	20										0
		n78	See CA_n78(2A) Bandwidth Combination 0 in Table 5.5A.2-1													
CA_n39A-n40A	CA_n39A-n40A	n39	5	10	15	20	25	30	40							0
		n40	5	10	15	20	25	30	40	50	60		80			
CA_n39A-n41A	CA_n39A-n41A	n39	5	10	15	20	25	30	40							0
		n41		10	15	20			40	50	60		80	90	100	
CA_n39A-n41C	CA_n39A-n41A	n39	5	10	15	20	25	30	40							0
		n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1													
CA_n39A-n41(2A)	CA_n39A-n41A	n39	5	10	15	20	25	30	40							0
		n41	See CA_n41(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													
CA_n39A-n79A	CA_n39A-n79A	n39	5	10	15	20	25	30	40							0
		n79							40	50	60		80		100	
CA_n40A-n41A	CA_n40A-n41A	n40	5	10	15	20	25	30	40	50	60		80			0
		n41		10	15	20			40	50	60		80	90	100	
		n40	5	10	15	20	25	30	40							1
		n41		10	15	20			40	50	60					
CA_n40A-n78A	CA_n40A-n78A	n40	5	10	15	20	25	30	40	50	60		80			0
		n78		10	15	20			40	50	60		80	90	100	
CA_n40A-n78(2A)	CA_n40A-n78A	n40	5	10	15	20	25	30	40	50	60		80			0

		n78	See CA_n78(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1												
CA_n40A-n79A	CA_n40A-n79A	n40	5	10	15	20	25	30	40	50	60		80		0
		n79							40	50	60		80	100	
		n40	5	10	15	20	25	30	40						1
		n79							40	50	60		80	100	
CA_n41A-n50A	CA_n41A-n50A	n41		10	15	20			40	50	60		80	90	0
		n50	5	10	15	20			40	50	60		80 ¹		
CA_n41A-n66A	CA_n41A-n66A	n41		10	15	20			40	50	60		80	90	0
		n66	5	10	15	20			40						
CA_n41(2A)-n66A	-	n41	See CA_n41(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1												0
		n66	5	10	15	20			40						
CA_n41C-n66A	-	n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1												0
		n66	5	10	15	20			40						
CA_n41A-n71A	CA_n41A-n71A	n41		10	15	20			40	50	60		80	90	0
		n71	5	10	15	20									
CA_n41A-n71B	-	n41		10	15	20		30	40	50	60		80	90	0
		n71	See CA_n71B Bandwidth Combination Set 0 in Table 5.5A.1-1												
CA_n41C-n71A	-	n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1												0
		n71	5	10	15	20									
CA_n41(2A)-n71A	-	n41	See CA_n41(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1												0
		n71	5	10	15	20									
CA_n41(2A)-n71B	-	n41	See CA_n41(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1												0
		n71	See CA_n71B Bandwidth Combination Set 0 in Table 5.5A.1-1												
CA_n41C-n71B	-	n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1												0
		n71	See CA_n71B Bandwidth Combination Set 0 in Table 5.5A.1-1												
CA_n41A-n78A	CA_n41A-n78A	n41		10	15	20			40	50	60		80	100	0
		n78		10	15	20			40	50	60		80	90	100
CA_n41A-n78A	CA_n41A-n78A	n41		10	15	20		30	40	50	60		80	90	1
		n78		10	15	20	25	30	40	50	60	70	80	90	100
CA_n41A-n79A	CA_n41A-n79A	n41		10	15	20			40	50	60		80	90	0
		n79							40	50	60		80	100	
		n41		10	15	20			40	50	60				1
		n79							40	50	60		80	100	
CA_n41C-n79A	CA_n41A-n79A CA_n41C	n41	See CA_n41C Bandwidth Combination Set 0 in Table 5.5A.1-1												0
		n79							40	50	60		80	100	

CA_n46A-n48A	CA_n46A-n48A	n46				20			40		60		80			0
		n48				20										
CA_n46B-n48A	CA_n46A-n48A	n46	See CA_n46B Bandwidth Combination Set 0 in 38.101-1 Table 5.5A.1-1													0
		n48				20										
CA_n46C-n48A	CA_n46A-n48A	n46	See CA_n46C Bandwidth Combination Set 0 in 38.101-1 Table 5.5A.1-1													0
		n48				20										
CA_n46D-n48A	CA_n46A-n48A	n46	See CA_n46D Bandwidth Combination Set 0 in 38.101-1 Table 5.5A.1-1													0
		n48				20										
CA_n46A-n66A	-	n46				20			40		60		80			0
		n66	5	10	15	20	25	30	40							
CA_n48A-n66A	CA_n48A-n66A	n48	5	10	15	20			40	50 ⁴	60 ⁴		80 ⁴	90 ⁴	100 ⁴	0
		n66	5	10	15	20			40							
CA_n48C-n66A	CA_n48A-n66A	n48	See CA_n48C Bandwidth Combination Set 0 in Table 5.5A.1-1													0
		n66	5	10	15	20			40							
CA_n48(2A)-n66A	CA_n48A-n66A	n48	See CA_n48(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n66	5	10	15	20			40							
CA_n50A-n78A	CA_n50A-n78A	n50	5	10	15	20		30	40	50	60		80 ¹			0
		n78		10	15	20			40	50	60		80	90	100	
CA_n66A-n70A	-	n66	5	10	15	20			40							0
		n70	5	10	15	20 ¹	25 ¹									
CA_n66B-n70A	-	n66	See CA_n66B Bandwidth Combination Set 0 in Table 5.5A.1-1													0
		n70	5	10	15	20 ¹	25 ¹									
CA_n66(2A)-n70A	-	n66	See CA_n66(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n70	5	10	15	20 ¹	25 ¹									
CA_n66A-n71A	CA_n66A-n71A	n66	5	10	15	20			40							0
		n71	5	10	15	20										
CA_n66(2A)-n71A	CA_n66A-n71A	n66	See CA_n66(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1													0
		n71	5	10	15	20										
CA_n66B-n71A	CA_n66A-n71A	n66	See CA_n66B Bandwidth Combination Set 0 in Table 5.5A.1-1													0
		n71	5	10	15	20										
CA_n66A-n77A	CA_n66A-n77A	n66	5	10	15	20			40							0
		n77		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n66A-n78A	CA_n66A-n78A	n66	5	10	15	20			40							0
		n78		10	15	20			40	50	60		80	90	100	
CA_n66A-n78(2A)	CA_n66A-n78A	n66	5	10	15	20		30	40							0

[illegible]

[illegible]

CA_n3A-n28A-n78A	-	n3	5	10	15	20									0
		n28	5	10	15	20 ²									
		n78		10	15	20			40	50	60	80	90	100	
CA_n3A-n28A-n78(2A)	-	n3	5	10	15	20									0
		n28	5	10	15	20 ²									
		n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A.2-1												
CA_n3A-n40A-n41A	CA_n3A-n40A CA_n3A-n41A CA_n40A-n41A	n3	5	10	15	20	25	30							0
		n40	5	10	15	20	25	30	40	50	60	80			
		n41		10	15	20			40	50	60	80	90	100	
CA_n3A-n41A-n79A	-	n3	5	10	15	20	25	30							0
		n41		10	15	20			40	50	60	80		100	
		n79							40	50	60	80		100	
		n3	5	10	15	20	25	30							1
		n41		10	15	20			40	50	60	80			
		n79							40	50	60	80		100	
CA_n5A-n66A-n78A	CA_n5A-n66A CA_n5A-n78A CA_n66A-n78A	n5	5	10	15	20									0
		n66	5	10	15	20	25	30	40						
		n78		10	15	20	25	30	40	50	60	80	90	100	
CA_n7A-n25A-n66A	CA_n7A-n25A CA_n7A-n66A CA_n25A-n66A	n7	5	10	15	20	25	30	40	50					0
		n25	5	10	15	20	25	30	40						
		n66	5	10	15	20	25	30	40						
CA_n7A-n28A-n78A	-	n7	5	10	15	20	25	30	40	50					0
		n28	5	10	15	20									
		n78		10	15	20	25	30	40	50	60	80	90	100	
CA_n7B-n28A-n78A	-	n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1												

		n28	5	10	15	20									
		n78		10	15	20	25	30	40	50	60	80	90	100	
CA_n7A-n66A-n78A	CA_n7A-n66A CA_n7A-n78A CA_n66A-n78A	n7	5	10	15	20	25	30	40	50					0
		n66	5	10	15	20	25	30	40						
		n78		10	15	20	25	30	40	50	60	80	90	100	
CA_n7A-n66A-n78(2A)	CA_n7A-n66A CA_n7A-n78A CA_n66A-n78A	n7	5	10	15	20	25	30	40	50					0
		n66	5	10	15	20	25	30	40						
		n78	See CA_n78(2A) Bandwidth Combination Set 1 in Table 5.5A.2-1												
CA_n8A-n39A-n41A	-	n8	5	10	15	20									0
		n39	5	10	15	20	25	30	40						
		n41		10	15	20			40	50	60	80		100	
		n8	5	10	15	20									1
		n39	5	10	15	20	25	30	40						
		n41		10	15	20			40	50	60				
CA_n8A-n41A-n79A	-	n8	5	10	15	20									0
		n41		10	15	20			40	50	60	80		100	
		n79							40	50	60	80		100	
		n8	5	10	15	20									1
		n41		10	15	20			40	50	60				
		n79							40	50	60	80		100	
CA_n20A-n28A-n78A	-	n20	5	10	15	20									0
		n28	5	10	15	20									
		n78		10	15	20		30	40	50	60	80	90	100	
CA_n25A-n41A-n66A	-	n25	5	10	15	20									0
		n41		10	15	20		30	40	50	60	80	90	100	
		n66	5	10	15	20			40						

[illegible]

CA_n1A-n3A-n7B-n28A	-	n1	5	10	15	20										0
		n3	5	10	15	20	25	30								
		n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													
		n28	5	10	15	20										
CA_n1A-n3A-n7A-n78A	-	n1	5	10	15	20										0
		n3	5	10	15	20	25	30								
		n7	5	10	15	20	25	30	40	50						
		n78		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n1A-n3A-n7B-n78A	-	n1	5	10	15	20										0
		n3	5	10	15	20	25	30								
		n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													
		n78		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n1A-n3A-n8A-n78A	-	n1	5	10	15	20										0
		n3	5	10	15	20	25	30								
		n8	5	10	15	20										
		n78		10	15	20			40	50	60		80	90 ¹	100	
CA_n1A-n3A-n28A-n78A	-	n1	5	10	15	20										0
		n3	5	10	15	20	25	30								
		n28	5	10	15	20 ²										
		n78		10	15	20			40	50	60		80	90 ¹	100	
CA_n3A-n7A-n28A-n78A	-	n3	5	10	15	20	25	30								0
		n7	5	10	15	20	25	30	40	50						
		n28	5	10	15	20										
		n78		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n3A-n7B-n28A-n78A	-	n3	5	10	15	20	25	30								0
		n7	See CA_n7B Bandwidth Combination Set 0 in Table 5.5A.1-1													
		n28	5	10	15	20										
		n78		10	15	20	25	30	40	50	60	70	80	90	100	
CA_n7A-n25A-n66A-n78A	-	n7	5	10	15	20	25	30	40	50						0
		n25	5	10	15	20	25	30	40							
		n66	5	10	15	20	25	30	40							

		n78		10	15	20	25	30	40	50	60	70	80	90	100	
NOTE 1: This UE channel bandwidth is optional in this release of the specification.																
NOTE 2: For the 20 MHz bandwidth, the minimum requirements are specified for NR UL carrier frequencies confined to either 713-723 MHz or 728-738 MHz.																
NOTE 3: The SCS of each channel bandwidth for NR band refers to Table 5.3.5-1.																

5.5B Configurations for DC

For an NR DC configuration specified in Table 5.5B-1, the bandwidth combination sets for the corresponding NR CA configuration in 5.5A.3, i.e. dual uplink inter-band carrier aggregation with uplink assigned to two NR bands, are applicable to Dual Connectivity.

Table 5.5B-1: Inter-band NR DC configurations (two bands)

NR DC configuration	Uplink NR DC configuration
DC_n2A-n5A	DC_n2A-n5A

5.5C Configurations for SUL

Table 5.5C-1: Supported channel bandwidths per SUL band combination

SUL configuration	NR Band	Channel bandwidth (MHz) (NOTE 1)												Bandwidth combination set
		5	10	15	20	25	30	40	50	60	80	90	100	
SUL_n41A-n80A	n41		10	15	20			40	50	60	80	90	100	0
	n80	5	10	15	20	25	30							
SUL_n41A-n81A	n41		10	15	20			40	50	60	80	90	100	0
	n81	5	10	15	20									
SUL_n41A-n95A	n41		10	15	20		30	40	50	60	80	90	100	0
	n95	5	10	15										
SUL_n77A-n80A	n77		10	15	20			40	50	60	80	90	100	0
	n80	5	10	15	20	25	30							
SUL_n77A-n84A	n77		10	15	20			40	50	60	80	90	100	0
	n84	5	10	15	20									
SUL_n78A-n80A	n78		10	15	20			40	50	60	80	90	100	0
	n80	5	10	15	20	25	30							
SUL_n78A-n81A	n78		10	15	20			40	50	60	80	90	100	0
	n81	5	10	15	20									
SUL_n78A-n82A	n78		10	15	20			40	50	60	80	90	100	0

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	n82	5	10	15	20									
SUL_n78A-n83A	n78		10	15	20			40	50	60	80	90	100	0
	n83	5	10	15	20									
SUL_n78A-n84A	n78		10	15	20			40	50	60	80	90	100	0
	n84	5	10	15	20									
SUL_n78A-n86A	n78		10	15	20			40	50	60	80	90	100	0
	n86	5	10	15	20									
SUL_n79A-n80A	n79							40	50	60	80		100	0
	n80	5	10	15	20	25	30							
SUL_n79A-n81A	n79							40	50	60	80		100	0
	n81	5	10	15	20									
SUL_n79A-n84A	n79							40	50	60	80		100	0
	n84	5	10	15	20									
SUL_n79A-n95A	n79							40	50	60	80		100	0
	n95	5	10	15										
NOTE 1: The SCS of each channel bandwidth for NR band refers to Table 5.3.5-1.														

Table 5.5C-2: Supported channel bandwidths per SUL band combination with intra-band non-contiguous CA

SUL band combination with intra-band non-contiguous CA	SUL configuration	NR Band	Channel bandwidth (MHz) (NOTE 1)												Bandwidth combination set
			5	10	15	20	25	30	40	50	60	80	90	100	
CA_n78(2A)-n86A	SUL_n78A-n86A	n78	See CA_n78(2A) Bandwidth Combination Set 0 in Table 5.5A. 2-1												0
		n86	5	10	15	20									
NOTE 1: The SCS of each channel bandwidth for NR band refers to Table 5.3.5-1.															